

CLASS 8 BIOLOGY FOUNDATION

Complete Exam-Ready Notes

Structured with Quick Summaries · Real-Life Examples · Points to Remember
Memory Hooks · Common Mistakes · Chapter-End Revision

Chapter 1: Crop Production and Management

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that plants make their own food and need water, air and sunlight.
- Basic difference between living and non-living things.
- Familiarity with common food items — cereals, pulses, vegetables and fruits.
- Awareness of the seasons of the year in India.

1.1 Crops and Agricultural Practices

 **Estimated time:** 12 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

When plants of the same kind are grown on a large scale at one place it is called a crop, and the series of steps a farmer follows from soil preparation to storage is called agricultural practices.

Real-Life Example

The wheat in your chapati and the rice in your lunch were both grown as crops — wheat in the cool rabi season and rice in the rainy kharif season.

Detailed Explanation

Definition

A crop is a large-scale cultivation of plants of the same kind in one field, and agricultural practices are the sequence of activities carried out by a farmer to raise that crop.

Key Idea

Indian crops are grouped by growing season. Kharif crops are sown in the rainy season (June–October) and need plenty of water — paddy, maize, soyabean, groundnut, cotton. Rabi crops are sown in winter (October–March) and need less water — wheat, gram, pea, mustard, linseed.

Working Principle

Seven basic practices are followed in order: preparation of soil, sowing, adding manure and fertilizers, irrigation, protection from weeds, harvesting, and storage. Skipping or delaying any step lowers the yield.

Applications

- Choosing kharif or rabi crops according to the local rainfall pattern.
- Crop rotation — growing a legume after a cereal to restore soil nitrogen.
- Planning a farming calendar so that sowing matches the arrival of the monsoon.

✓ Points to Remember (Exam Focus)

- ✓ Crop = plants of the same kind grown on a large scale at one place.
- ✓ Kharif = sown with the monsoon (June–Oct); Rabi = sown in winter (Oct–Mar).
- ✓ Paddy, maize, cotton, groundnut and soyabean are kharif crops.
- ✓ Wheat, gram, pea, mustard and linseed are rabi crops.
- ✓ The seven agricultural practices always follow the same order, ending with storage.

⚠ Common Student Mistakes

- ⚠ Calling paddy a rabi crop — paddy needs standing water and is a kharif crop.
- ⚠ Thinking 'crop' means any group of plants; it must be plants of the same kind grown on a large scale.
- ⚠ Mixing up the order of practices — irrigation comes after sowing and manuring, not before soil preparation.



Memory Hook

'KhaRif comes with the Rain, Rabi Rests in the cold' — and remember the seven steps as 'Soil, Sow, Manure, Water, Weed, Harvest, Store'.



Advanced Insights

The kharif/rabi division is really a water-availability division. A third group, zaid crops (watermelon, cucumber, muskmelon), is grown in the short summer gap between rabi harvest and kharif sowing wherever irrigation is assured.

1.2 Preparation of Soil and Sowing of Seeds

🕒 **Estimated time:** 14 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★



Quick Summary

Loosening and levelling the soil by ploughing lets roots breathe and grow deep, and sowing places good-quality seed at the right depth and spacing using a seed drill or traditional funnel.



Real-Life Example

A gardener always digs and turns the soil in a flowerpot before planting seeds — that is ploughing on a small scale.



Detailed Explanation

Definition

Preparation of soil is the loosening, turning and levelling of soil before sowing; sowing is the placing of seeds in the prepared soil at a proper depth and distance.

Key Idea

Loose soil holds more air and water, lets earthworms and microbes work freely, and allows roots to penetrate deeply. Nutrient-rich soil is also brought up to the surface when the soil is turned.

Working Principle

Ploughing (with a plough, hoe or cultivator) loosens the soil, levelling with a leveller stops water and soil from being washed away, and sowing is done either by broadcasting by hand, with a funnel, or with a seed drill attached to a tractor. Before sowing, damaged seeds are removed by placing seeds in water — damaged seeds are hollow and float.

Applications

- Seed drills sow seeds at uniform depth and cover them, so birds cannot pick them up.
- Transplantation is used for paddy, tomato and chilli — seedlings are raised in a nursery and then moved to the field.
- The float test in salt water separates healthy seeds from damaged ones before sowing.

✓ Points to Remember (Exam Focus)

- ✓ Ploughing loosens the soil, allows aeration, and brings nutrient-rich soil to the top.
- ✓ Tools used: plough, hoe, cultivator (for ploughing); leveller (for levelling).
- ✓ Damaged seeds float in water because they are hollow and lighter.
- ✓ A seed drill sows seeds at the correct depth and spacing and saves time and seed.
- ✓ Correct spacing prevents overcrowding and competition for sunlight, water and nutrients.

⚠ Common Student Mistakes

- ⚠ Saying that damaged seeds sink — they float because they are hollow.
- ⚠ Confusing a cultivator (ploughing with a tractor) with a seed drill (sowing with a tractor).
- ⚠ Assuming deeper sowing is always better — seeds sown too deep cannot reach the surface and die.

🧠 Memory Hook

'Loose soil = Long roots'; and 'Floaters are Failures' — the seeds that float are the damaged ones.

🚀 Advanced Insights

Zero-tillage farming skips ploughing altogether and sows directly into the residue of the previous crop. It conserves soil moisture and organic carbon, and is increasingly used for wheat sown after paddy in north India.

II Quick Recap — Topics Covered So Far

- Crops and Agricultural Practices — When plants of the same kind are grown on a large scale at one place it is called a crop, and the series of steps a farmer follows from soil preparation to storage is called agricultural practices.
- Preparation of Soil and Sowing of Seeds — Loosening and levelling the soil by ploughing lets roots breathe and grow deep, and sowing places good-quality seed at the right depth and spacing using a seed drill or traditional funnel.

1.3 Manure, Fertilizers and Irrigation

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Manure and fertilizers replace the nutrients a crop removes from the soil, while irrigation supplies water at the right time — both decide how much the field finally yields.

Real-Life Example

The compost pit in a housing society converts kitchen waste into manure, while the drip pipes in a modern orchard deliver water drop by drop straight to each root.

Detailed Explanation

Definition

Manure is a natural organic substance obtained by decomposing plant and animal wastes; a fertilizer is an inorganic salt manufactured in a factory that supplies a specific plant nutrient. Irrigation is the supply of water to crops at regular intervals.

Key Idea

Manure adds humus, improves soil texture and water-holding capacity, but supplies few nutrients. Fertilizers are nutrient-rich and act fast, but do not add humus, and excess use damages soil fertility and pollutes water.

Working Principle

Nutrients removed by a crop must be replenished — otherwise the soil becomes deficient. Water is needed because nutrients are absorbed by roots only in dissolved form, and water is a raw material for photosynthesis and for germination.

Applications

- Traditional irrigation: moat (pulley system), chain pump, dhekli and rahat.
- Modern irrigation: sprinkler system for sandy soils and uneven land; drip system for fruit trees and gardens.
- Crop rotation with leguminous plants such as gram and pea fixes nitrogen and reduces the need for fertilizer.

✓ Points to Remember (Exam Focus)

- ✓ Manure = organic, adds humus, slow-acting; Fertilizer = inorganic, nutrient-specific, fast-acting.
- ✓ Urea, ammonium sulphate, superphosphate and NPK are common fertilizers.
- ✓ Excessive fertilizer use destroys soil fertility and pollutes water bodies.
- ✓ Drip irrigation is the best method in regions with a scarcity of water.
- ✓ Rhizobium bacteria in the root nodules of leguminous plants fix atmospheric nitrogen.

⚠ Common Student Mistakes

- ⚠ Using 'manure' and 'fertilizer' as the same word — manure is natural and fertilizer is factory-made.
- ⚠ Thinking more fertilizer always means more yield; over-use makes the soil infertile.
- ⚠ Forgetting that sprinkler irrigation, not drip, is the method suited to sandy and uneven land.

**Memory Hook**

'MANure comes from Man and animals (natural); FERTilizer comes from a FactoRy' — and 'Drip = Drop by drop saves the crop'.

**Advanced Insights**

Excess nitrogen fertilizer washed into ponds causes eutrophication: algae bloom, then die and decay, and the decomposers consume the dissolved oxygen — killing fish. This links crop management directly to water pollution.

1.4 Weeding, Harvesting and Storage

🕒 **Estimated time:** 13 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Weeds are removed so they do not steal nutrients, the ripened crop is cut at harvest, grain is separated by threshing and winnowing, and the dried grain is stored safely against moisture and pests.

**Real-Life Example**

The 'Kanaji' or 'khalihan' where villagers thresh paddy, and the huge cylindrical silos beside railway lines used by the Food Corporation of India, are the harvesting and storage steps in action.

**Detailed Explanation****Definition**

Weeds are unwanted plants growing naturally along with a crop; harvesting is the cutting of the mature crop; threshing separates grain from the chaff; winnowing separates lighter chaff from heavier grain using wind.

Key Idea

Weeds compete with the crop for water, nutrients, space and light, so removing them directly raises yield. After harvesting, grain must be dried in sunlight to below about 14% moisture before storage, otherwise fungi and pests destroy it.

Working Principle

Weeding is done manually (khurpi, seed drill before sowing, uprooting) or chemically with weedicides such as 2,4-D. Harvesting is done with a sickle or a harvester; a combine does harvesting and threshing together. Grain is then stored in jute bags, metal bins, granaries or silos, with dried neem leaves or fumigants to keep pests away.

Applications

- Weedicides are sprayed on fields before flowering to kill weeds without harming the crop.
- A combine harvester saves labour by cutting and threshing in a single operation.
- Farmers use dried neem leaves in home storage and fumigants in large silos and granaries.

✓ Points to Remember (Exam Focus)

- ✓ Weeds compete for nutrients, water, space and sunlight, and reduce yield.
- ✓ 2,4-D is a common weedicide; weedicides should be sprayed with the face covered.
- ✓ Threshing = grain separated from stalk; Winnowing = chaff blown away by wind.
- ✓ Grain must be dried in sunlight before storage to prevent attack by fungi and pests.
- ✓ A combine performs both harvesting and threshing.

⚠ Common Student Mistakes

- ⚠ Swapping threshing and winnowing — threshing beats out the grain, winnowing blows away the chaff.
- ⚠ Believing weedicides are harmless; they can affect the health of the person spraying them.
- ⚠ Storing grain without drying it, which is the main cause of spoilage by microorganisms.



Memory Hook

'THresh to THump the grain out, WINnow with the WIND' — and 'Dry it before you Store it'.



Advanced Insights

The Green Revolution of the 1960s combined high-yielding varieties, fertilizers and assured irrigation to make India self-sufficient in food grain. Its later cost — falling water tables and declining soil fertility — is why sustainable practices such as crop rotation are being revived.

II Quick Recap — Topics Covered So Far

- **Manure, Fertilizers and Irrigation** — Manure and fertilizers replace the nutrients a crop removes from the soil, while irrigation supplies water at the right time — both decide how much the field finally yields.
- **Weeding, Harvesting and Storage** — Weeds are removed so they do not steal nutrients, the ripened crop is cut at harvest, grain is separated by threshing and winnowing, and the dried grain is stored safely against moisture and pests.

Chapter 1 — End of Chapter Revision

All Memory Hooks

- Crops and Agricultural Practices: 'KhaRif comes with the Rain, Rabi Rests in the cold' — and remember the seven steps as 'Soil, Sow, Manure, Water, Weed, Harvest, Store'.
- Preparation of Soil and Sowing of Seeds: 'Loose soil = Long roots'; and 'Floaters are Failures' — the seeds that float are the damaged ones.
- Manure, Fertilizers and Irrigation: 'MANure comes from Man and animals (natural); FERTilizer comes from a FactoRy' — and 'Drip = Drop by drop saves the crop'.
- Weeding, Harvesting and Storage: 'Thresh to THump the grain out, WINnow with the WIND' — and 'Dry it before you Store it'.

All Common Mistakes

-  Calling paddy a rabi crop — paddy needs standing water and is a kharif crop.
-  Thinking 'crop' means any group of plants; it must be plants of the same kind grown on a large scale.
-  Mixing up the order of practices — irrigation comes after sowing and manuring, not before soil preparation.
-  Saying that damaged seeds sink — they float because they are hollow.
-  Confusing a cultivator (ploughing with a tractor) with a seed drill (sowing with a tractor).
-  Assuming deeper sowing is always better — seeds sown too deep cannot reach the surface and die.
-  Using 'manure' and 'fertilizer' as the same word — manure is natural and fertilizer is factory-made.
-  Thinking more fertilizer always means more yield; over-use makes the soil infertile.
-  Forgetting that sprinkler irrigation, not drip, is the method suited to sandy and uneven land.
-  Swapping threshing and winnowing — threshing beats out the grain, winnowing blows away the chaff.
-  Believing weedicides are harmless; they can affect the health of the person spraying them.
-  Storing grain without drying it, which is the main cause of spoilage by microorganisms.

Quick Revision Section

- Crops and Agricultural Practices: When plants of the same kind are grown on a large scale at one place it is called a crop, and the series of steps a farmer follows from soil preparation to storage is called agricultural practices.
- Preparation of Soil and Sowing of Seeds: Loosening and levelling the soil by ploughing lets roots breathe and grow deep, and sowing places good-quality seed at the right depth and spacing using a seed drill or traditional funnel.

- **Manure, Fertilizers and Irrigation:** Manure and fertilizers replace the nutrients a crop removes from the soil, while irrigation supplies water at the right time — both decide how much the field finally yields.
- **Weeding, Harvesting and Storage:** Weeds are removed so they do not steal nutrients, the ripened crop is cut at harvest, grain is separated by threshing and winnowing, and the dried grain is stored safely against moisture and pests.

Chapter 2: Micro-organisms

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that living things are made of cells.
- Knowledge that some diseases spread from one person to another.
- Familiarity with a simple microscope and why it is needed.
- Basic understanding that food gets spoiled if left in the open.

2.1 Microorganisms and Their Classification

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Microorganisms are living organisms too small to be seen with the naked eye, and they are grouped into five classes — bacteria, fungi, protozoa, algae and viruses.

Real-Life Example

The greenish slime on a wet wall is algae, the white fuzz on stale bread is fungus, and the curd in your kitchen is made by bacteria — three different microbes in one house.

Detailed Explanation

Definition

Microorganisms, or microbes, are organisms so small that they cannot be seen with the unaided eye and need a microscope to be observed.

Key Idea

Microbes may be unicellular (bacteria, some algae, protozoa) or multicellular (most fungi, some algae). They live everywhere — in soil, water, air, inside our bodies, in hot springs, in ice and in deserts.

Working Principle

Bacteria are classified by shape into cocci (spherical), bacilli (rod-shaped), spirilla (spiral) and vibrio (comma-shaped). Gram staining separates bacteria into Gram-positive (thick peptidoglycan wall, stain purple) and Gram-negative (thin wall, stain pink).

Applications

- Amoeba, Paramecium and Plasmodium are protozoa; Chlamydomonas and Spirogyra are algae.
- Yeast, Penicillium, Aspergillus, Rhizopus and mushrooms are fungi.
- Lichen is a symbiotic partnership of an alga and a fungus; mycorrhiza is a fungus–root partnership.

✓ Points to Remember (Exam Focus)

- ✓ Five groups of microbes: bacteria, fungi, protozoa, algae and viruses.
- ✓ Bacterial shapes: coccus (round), bacillus (rod), spirillum (spiral), vibrio (comma).
- ✓ Cyanobacteria (blue-green algae) are bacteria that photosynthesise and fix nitrogen.
- ✓ Microbes are found in ice, hot springs, deserts and marshy land — everywhere.
- ✓ Antonie van Leeuwenhoek first observed microorganisms in 1674.

⚠ Common Student Mistakes

- ⚠ Calling cyanobacteria 'algae' — they are bacteria (prokaryotes), despite the old name blue-green algae.
- ⚠ Assuming all microorganisms are unicellular — many fungi and algae are multicellular.
- ⚠ Thinking a microscope is only needed for bacteria; almost all microbes need magnification.



Memory Hook

'Ba-Fu-Pro-Al-Vi' — Bacteria, Fungi, Protozoa, Algae, Viruses; and for shapes, 'Cocci are Circles, Bacilli are Bars'.



Advanced Insights

Extremophiles are microbes that thrive where nothing else can — thermophiles in hot springs above 80 °C, halophiles in salt lakes. The heat-stable enzyme Taq polymerase, taken from a hot-spring bacterium, made modern DNA testing possible.

2.2 Viruses — Living or Non-living?

🕒 **Estimated time:** 10 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

A virus is on the borderline between the living and the non-living — it reproduces only inside a living host cell and is completely inert outside it.



Real-Life Example

The common cold, influenza, chickenpox, polio and COVID-19 are all caused by viruses, which is why antibiotics do not help against them.



Detailed Explanation

Definition

A virus is an infectious particle made of a nucleic acid core (DNA or RNA) surrounded by a protein coat called a capsid, which multiplies only inside a living host cell.

Key Idea

Outside a host, a virus can be crystallised like a chemical and shows no sign of life. Inside a host cell, it hijacks the cell's machinery to make copies of itself — so it is described as an obligate intracellular parasite.

Working Principle

The virus attaches to a specific host cell, injects its nucleic acid, forces the cell to manufacture viral parts, assembles new virus particles and finally bursts the cell to release them.

Applications

- Bacteriophages are viruses that attack bacteria and are used in phage therapy.
- Vaccines use weakened or inactivated viruses to train the immune system.
- Viral diseases in humans: polio, measles, chickenpox, mumps, influenza, AIDS, dengue.

✓ Points to Remember (Exam Focus)

- ✓ Virus = nucleic acid (DNA or RNA) + protein coat; never both DNA and RNA.
- ✓ Living characteristic: it reproduces and has genetic material.
- ✓ Non-living characteristics: it can be crystallised, has no cell structure and no metabolism of its own.
- ✓ Antibiotics do NOT work against viruses — only vaccines and antiviral drugs help.
- ✓ Viruses are much smaller than bacteria and are seen only with an electron microscope.

⚠ Common Student Mistakes

- ⚠ Taking antibiotics for a cold or flu — these are viral and antibiotics have no effect.
- ⚠ Saying a virus contains both DNA and RNA; it contains only one of the two.
- ⚠ Calling viruses cells — they have no cell membrane, cytoplasm or organelles.

🧠 Memory Hook

'A virus is a pirate ship — dead in the harbour, deadly once it boards a living cell.'

🚀 Advanced Insights

Some viruses insert their genes permanently into the host chromosome; about 8% of the human genome is made of such ancient viral sequences, and a few of them now perform useful functions such as helping form the placenta.

II Quick Recap — Topics Covered So Far

- Microorganisms and Their Classification — Microorganisms are living organisms too small to be seen with the naked eye, and they are grouped into five classes — bacteria, fungi, protozoa, algae and viruses.
- Viruses — Living or Non-living? — A virus is on the borderline between the living and the non-living — it reproduces only inside a living host cell and is completely inert outside it.

2.3 Useful Microorganisms

🕒 **Estimated time:** 14 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Microbes are used to make curd, bread and alcohol, to produce antibiotics and vaccines, to clean up the environment and to increase soil fertility.

Real-Life Example

The bubbles in a rising dough, the sourness of curd, and the penicillin in a prescription are all the work of useful microorganisms.

Detailed Explanation

Definition

Useful microorganisms are microbes whose activities benefit humans — in food production, medicine, agriculture and waste management.

Key Idea

The single most important microbial process for humans is fermentation — the breakdown of sugar into alcohol and carbon dioxide (or into acid) in the absence of oxygen, discovered by Louis Pasteur in 1857.

Working Principle

Lactobacillus converts milk sugar to lactic acid, curdling milk into curd. Yeast (*Saccharomyces*) ferments sugar to alcohol and CO₂, making bread rise and producing wine and beer. Soil bacteria decompose dead matter and return nutrients to the soil.

Applications

- Antibiotics: penicillin from the fungus *Penicillium* (Alexander Fleming, 1929), streptomycin, tetracycline, erythromycin.
- Vaccines against cholera, tuberculosis, smallpox and hepatitis (Edward Jenner, smallpox, 1798).
- Rhizobium in legume root nodules and Azotobacter in soil fix atmospheric nitrogen; blue-green algae act as biofertilizers in paddy fields.

✓ **Points to Remember (Exam Focus)**

- ✓ Lactobacillus → curd; Yeast → bread, wine, beer (fermentation).

- ✓ Penicillin was the first antibiotic, discovered by Alexander Fleming from *Penicillium*.
- ✓ A vaccine contains dead or weakened microbes that make the body produce antibodies.
- ✓ *Rhizobium* lives in root nodules of leguminous plants and fixes nitrogen.
- ✓ Decomposers convert dead plant and animal remains into humus.

⚠ Common Student Mistakes

- ⚠ Saying yeast makes curd — curd is made by *Lactobacillus*; yeast makes bread and alcohol.
- ⚠ Confusing antibiotic with vaccine: an antibiotic kills bacteria already inside you, a vaccine prevents the disease beforehand.
- ⚠ Taking antibiotics without completing the prescribed course, which breeds resistant bacteria.



Memory Hook

'Lacto for Lassi, Yeast for Yeast-bread' — and 'Fleming Found Fungus's Penicillin'.



Advanced Insights

Antibiotic resistance arises by natural selection: a few bacteria carrying a resistance gene survive treatment and multiply. This is why doctors insist on finishing the full course — stopping early leaves behind exactly the toughest survivors.

2.4 Harmful Microorganisms and Food Preservation

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

Pathogens cause communicable diseases in humans, animals and plants, and spoil food — which is why food is preserved by heat, cold, chemicals, drying or salting.



Real-Life Example

Milk in a sealed tetra pack lasts for months because it has been pasteurised, and pickles keep for a year because oil and salt keep microbes out.



Detailed Explanation

Definition

Disease-causing microorganisms are called pathogens; a disease that spreads from an infected person to a healthy person is a communicable disease. Food preservation is the treatment of food to prevent microbial spoilage.

Key Idea

Pathogens spread through air (cough, sneeze), water (contaminated drinking water), food, direct contact, and through carriers — animals such as the female *Anopheles* mosquito (malaria), the *Aedes* mosquito (dengue) and the housefly.

Working Principle

Preservation works by removing what microbes need: heat kills them (pasteurisation, boiling, canning), cold slows them (refrigeration, freezing), drying and salting remove water, and chemical preservatives, sugar and oil make the food unsuitable for growth.

Applications

- Human diseases: tuberculosis, cholera, typhoid (bacteria); malaria, kala-azar (protozoa); polio, chickenpox, influenza (viruses).
- Plant diseases: citrus canker (bacteria), rust of wheat (fungi), yellow vein mosaic of bhindi (virus).
- Preservatives: sodium benzoate and sodium metabisulphite in jams, squashes and sauces.

✓ Points to Remember (Exam Focus)

- ✓ Pasteurisation: milk heated to about 70 °C for 15–30 seconds, then chilled suddenly.
- ✓ Female *Anopheles* mosquito carries the malarial parasite *Plasmodium*.
- ✓ Salt and sugar preserve food by drawing water out of microbial cells (osmosis).
- ✓ Common preservatives: sodium benzoate, sodium metabisulphite; oil and vinegar in pickles.
- ✓ Some fungi produce toxins; food poisoning is caused by eating food spoiled by microbes.

⚠ Common Student Mistakes

- ⚠ Thinking pasteurisation sterilises milk completely — it only kills most microbes, so milk must still be refrigerated.
- ⚠ Confusing the vector with the pathogen: the mosquito is the carrier, *Plasmodium* is the pathogen.
- ⚠ Believing refrigeration kills microorganisms — it only slows their growth down.



Memory Hook

'Pasteur Pasteurises' — and to preserve, 'Too Hot, Too Cold, Too Dry or Too Salty for a microbe to live'.



Advanced Insights

The nitrogen cycle links harmful and helpful microbes: nitrogen-fixing bacteria convert N_2 to nitrates, nitrifying bacteria oxidise ammonia, and denitrifying bacteria return nitrogen to the atmosphere — keeping the atmospheric level of nitrogen roughly constant at 78%.

II Quick Recap — Topics Covered So Far

- Useful Microorganisms — Microbes are used to make curd, bread and alcohol, to produce antibiotics and vaccines, to clean up the environment and to increase soil fertility.
- Harmful Microorganisms and Food Preservation — Pathogens cause communicable diseases in humans, animals and plants, and spoil food — which is why food is preserved by heat, cold, chemicals, drying or salting.

Chapter 2 — End of Chapter Revision

All Memory Hooks

- Microorganisms and Their Classification: 'Ba-Fu-Pro-Al-Vi' — Bacteria, Fungi, Protozoa, Algae, Viruses; and for shapes, 'Cocci are Circles, Bacilli are Bars'.
- Viruses — Living or Non-living?: 'A virus is a pirate ship — dead in the harbour, deadly once it boards a living cell.'
- Useful Microorganisms: 'Lacto for Lassi, Yeast for Yeast-bread' — and 'Fleming Found Fungus's Penicillin'.
- Harmful Microorganisms and Food Preservation: 'Pasteur Pasteurises' — and to preserve, 'Too Hot, Too Cold, Too Dry or Too Salty for a microbe to live'.

All Common Mistakes

- ⚠ Calling cyanobacteria 'algae' — they are bacteria (prokaryotes), despite the old name blue-green algae.
- ⚠ Assuming all microorganisms are unicellular — many fungi and algae are multicellular.
- ⚠ Thinking a microscope is only needed for bacteria; almost all microbes need magnification.
- ⚠ Taking antibiotics for a cold or flu — these are viral and antibiotics have no effect.
- ⚠ Saying a virus contains both DNA and RNA; it contains only one of the two.
- ⚠ Calling viruses cells — they have no cell membrane, cytoplasm or organelles.
- ⚠ Saying yeast makes curd — curd is made by Lactobacillus; yeast makes bread and alcohol.
- ⚠ Confusing antibiotic with vaccine: an antibiotic kills bacteria already inside you, a vaccine prevents the disease beforehand.
- ⚠ Taking antibiotics without completing the prescribed course, which breeds resistant bacteria.
- ⚠ Thinking pasteurisation sterilises milk completely — it only kills most microbes, so milk must still be refrigerated.
- ⚠ Confusing the vector with the pathogen: the mosquito is the carrier, Plasmodium is the pathogen.
- ⚠ Believing refrigeration kills microorganisms — it only slows their growth down.

Quick Revision Section

- Microorganisms and Their Classification: Microorganisms are living organisms too small to be seen with the naked eye, and they are grouped into five classes — bacteria, fungi, protozoa, algae and viruses.
- Viruses — Living or Non-living?: A virus is on the borderline between the living and the non-living — it reproduces only inside a living host cell and is completely inert outside it.

- **Useful Microorganisms:** Microbes are used to make curd, bread and alcohol, to produce antibiotics and vaccines, to clean up the environment and to increase soil fertility.
- **Harmful Microorganisms and Food Preservation:** Pathogens cause communicable diseases in humans, animals and plants, and spoil food — which is why food is preserved by heat, cold, chemicals, drying or salting.

Chapter 3: Conservation of Plants and Animals

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that plants and animals depend on one another for food.
- Basic understanding of a forest as a habitat.
- Awareness that some animals such as the tiger are rare.
- Knowledge that trees are cut for wood and paper.

3.1 Biodiversity and Deforestation

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Biodiversity is the variety of living organisms in an area, and deforestation — the clearing of forests for other uses — is the biggest threat to it.

Real-Life Example

Landslides in hilly regions after the hill slopes are cleared of trees, and the dust storms over deforested plains, are deforestation showing its effects within a few years.

Detailed Explanation

Definition

Biodiversity is the variety of plants, animals and microorganisms found in an area, together with the ecosystems they form. Deforestation is the clearing of forests and using that land for other purposes.

Key Idea

Forests hold the soil with their roots, recycle water through transpiration, absorb carbon dioxide and release oxygen. Removing them therefore causes soil erosion, floods, droughts, global warming and loss of habitat all at once.

Working Principle

Natural causes of deforestation include forest fires and severe droughts. Human causes — the larger share — are cutting trees for agriculture, houses, factories, roads, furniture and firewood.

Applications

- Reforestation — replanting the same species that were cut — restores a degraded forest.
- The Van Mahotsav programme plants trees across India every July.
- Recycling paper: recycling one tonne of paper saves about 17 mature trees.

✓ Points to Remember (Exam Focus)

- ✓ Consequences of deforestation: soil erosion, floods, droughts, desertification, global warming.
- ✓ Fewer trees → less CO₂ absorbed → more greenhouse effect → rise in earth's temperature.
- ✓ Deforestation lowers the water-holding capacity of soil, disturbing the water cycle.
- ✓ The top layer of soil is exposed and gradually converts fertile land into desert.
- ✓ Paper can be recycled about 5–7 times; saving paper saves trees.

⚠ Common Student Mistakes

- ⚠ Thinking deforestation only means fewer trees — its main damage is soil erosion and climate change.
- ⚠ Confusing reforestation (replanting where forest was cut) with afforestation (planting on new land).
- ⚠ Assuming forest fires are always man-made; lightning and extreme heat also start them.

**Memory Hook**

'No Trees → No Roots → No Soil → No Rain' — the four dominoes of deforestation.

**Advanced Insights**

Forests are carbon sinks: a mature tropical forest locks up several hundred tonnes of carbon per hectare. When burnt, that carbon returns to the atmosphere as CO₂, which is why deforestation accounts for roughly a tenth of global greenhouse-gas emissions.

3.2 Wildlife Conservation and Protected Areas

🕒 **Estimated time:** 13 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Wildlife is protected in wildlife sanctuaries, national parks and biosphere reserves, which conserve species in their own natural habitat.

**Real-Life Example**

Kaziranga National Park in Assam saved the one-horned rhinoceros, and Project Tiger raised India's tiger numbers from a few hundred to over three thousand.

**Detailed Explanation****Definition**

A wildlife sanctuary is an area where animals are protected from poaching and disturbance; a national park is a larger reserve preserving the whole ecosystem including flora, fauna and landscape; a biosphere reserve is a very large protected area that conserves biodiversity along with the culture of the people living there.

Key Idea

Conservation in the natural habitat (in-situ) preserves not just the animal but the entire web of relationships it depends on. Zoos and seed banks are ex-situ conservation and are only a supporting measure.

Working Principle

Endangered species are identified and listed in the Red Data Book maintained by the IUCN. Their habitat is then protected by law, poaching is banned, and breeding programmes are run where numbers are critically low.

Applications

- Pachmarhi Biosphere Reserve contains Satpura National Park and Bori and Pachmarhi wildlife sanctuaries.
- Project Tiger was launched in 1973 to protect the tiger, India's national animal.
- Migratory birds such as the Siberian crane travel thousands of kilometres to Indian wetlands each winter.

✓ Points to Remember (Exam Focus)

- ✓ Endemic species: found only in a particular area, e.g. sal and wild mango in Pachmarhi.
- ✓ Endangered species: numbers so low that they are at risk of extinction, e.g. tiger, rhino, great Indian bustard.
- ✓ Extinct species: no longer found anywhere on earth, e.g. dodo, Indian cheetah.
- ✓ The Red Data Book records all endangered plants and animals.
- ✓ A biosphere reserve is the largest protected area and may contain national parks and sanctuaries within it.

⚠ Common Student Mistakes

- ⚠ Swapping sanctuary and national park — a sanctuary protects animals, a national park protects the whole ecosystem.
- ⚠ Confusing 'endemic' (found only there) with 'endangered' (at risk of extinction).
- ⚠ Thinking migration is aimless wandering; it is a seasonal, purposeful journey for breeding or climate.

Memory Hook

'Sanctuary Saves animals, National park saves Nature, Biosphere is the Biggest' — $S < N < B$ in size and scope.

Advanced Insights

Habitat fragmentation can be as damaging as habitat loss. When a road or canal cuts a forest into pieces, small isolated populations lose genetic diversity and become inbred, so wildlife corridors are now built to reconnect fragments.

II Quick Recap — Topics Covered So Far

- **Biodiversity and Deforestation** — Biodiversity is the variety of living organisms in an area, and deforestation — the clearing of forests for other uses — is the biggest threat to it.
- **Wildlife Conservation and Protected Areas** — Wildlife is protected in wildlife sanctuaries, national parks and biosphere reserves, which conserve species in their own natural habitat.

3.3 Recycling of Paper and Sustainable Practices

 **Estimated time:** 10 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★ ★

Quick Summary

Reducing, reusing and recycling paper directly saves trees, energy and water, and is the simplest conservation step a student can take.

Real-Life Example

The blank backs of used answer sheets, old newspapers sold to the kabadiwala, and recycled-paper notebooks all keep trees standing.

Detailed Explanation

Definition

Recycling of paper is the process of reprocessing used paper into pulp and making fresh paper from it, instead of using fresh wood pulp.

Key Idea

Paper is made from wood pulp, so every tonne of paper saved is roughly seventeen trees left standing, along with large savings in water and electricity.

Working Principle

Used paper is collected, soaked and pulped, ink is removed, and the clean pulp is spread, pressed and dried into new sheets. Because fibres shorten each time, paper can be recycled about five to seven times.

Applications

- Using both sides of a sheet before discarding it.
- Collecting old newspapers and notebooks for recycling instead of burning them.
- Buying recycled-paper stationery and avoiding unnecessary printing.

✓ Points to Remember (Exam Focus)

- ✓ 1 tonne of recycled paper saves about 17 mature trees.
- ✓ Paper can be recycled about 5–7 times before the fibres become too short.
- ✓ The three R's are Reduce, Reuse and Recycle — in that order of priority.
- ✓ Burning paper adds carbon dioxide and pollutants to the air; recycling avoids this.
- ✓ Van Mahotsav is a tree-planting festival celebrated across India.

⚠ Common Student Mistakes

- ⚠ Putting 'Recycle' first among the three R's — reducing use comes first.
- ⚠ Thinking all paper is recyclable; waxed, laminated and heavily soiled paper is not.
- ⚠ Assuming planting saplings immediately replaces a felled tree — a mature tree takes decades.



Memory Hook

'Reduce, Reuse, Recycle — in that order, R1 before R3.'



Advanced Insights

Every recycled sheet also avoids the methane released when paper decomposes in a landfill. Because methane traps roughly 25 times more heat than CO₂, paper recycling has a climate benefit far larger than the trees alone would suggest.

II Quick Recap — Topics Covered So Far

- Recycling of Paper and Sustainable Practices — Reducing, reusing and recycling paper directly saves trees, energy and water, and is the simplest conservation step a student can take.

Chapter 3 — End of Chapter Revision

All Memory Hooks

- Biodiversity and Deforestation: 'No Trees → No Roots → No Soil → No Rain' — the four dominoes of deforestation.
- Wildlife Conservation and Protected Areas: 'Sanctuary Saves animals, National park saves Nature, Biosphere is the Biggest' — $S < N < B$ in size and scope.
- Recycling of Paper and Sustainable Practices: 'Reduce, Reuse, Recycle — in that order, R1 before R3.'

All Common Mistakes

-  Thinking deforestation only means fewer trees — its main damage is soil erosion and climate change.
-  Confusing reforestation (replanting where forest was cut) with afforestation (planting on new land).
-  Assuming forest fires are always man-made; lightning and extreme heat also start them.
-  Swapping sanctuary and national park — a sanctuary protects animals, a national park protects the whole ecosystem.
-  Confusing 'endemic' (found only there) with 'endangered' (at risk of extinction).
-  Thinking migration is aimless wandering; it is a seasonal, purposeful journey for breeding or climate.
-  Putting 'Recycle' first among the three R's — reducing use comes first.
-  Thinking all paper is recyclable; waxed, laminated and heavily soiled paper is not.
-  Assuming planting saplings immediately replaces a felled tree — a mature tree takes decades.

Quick Revision Section

- Biodiversity and Deforestation: Biodiversity is the variety of living organisms in an area, and deforestation — the clearing of forests for other uses — is the biggest threat to it.
- Wildlife Conservation and Protected Areas: Wildlife is protected in wildlife sanctuaries, national parks and biosphere reserves, which conserve species in their own natural habitat.
- Recycling of Paper and Sustainable Practices: Reducing, reusing and recycling paper directly saves trees, energy and water, and is the simplest conservation step a student can take.

Chapter 4: Cell — Structure and Functions

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that all living things are built from smaller units.
- Familiarity with a compound microscope and how to focus it.
- Basic difference between plants and animals.
- Understanding that living things grow, respire and reproduce.

4.1 Discovery of the Cell and Cell Theory

 **Estimated time:** 11 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Robert Hooke first saw dead cork cells in 1665, and the cell theory that followed states that all living organisms are made of cells and that every cell comes from a pre-existing cell.

Real-Life Example

A slice of onion peel under a school microscope shows neat brick-like cells — the same view that began the study of biology as we know it.

Detailed Explanation

Definition

A cell is the basic structural and functional unit of life — the smallest unit that can carry out all the activities of living.

Key Idea

Cell theory has three parts: all organisms are composed of cells; the cell is the basic unit of structure and function in living things; and all cells arise from pre-existing cells (Rudolf Virchow).

Working Principle

Hooke examined a thin slice of cork under his own microscope and saw empty compartments which reminded him of the small rooms in a monastery — 'cellulae' — and named them cells. Leeuwenhoek later saw the first living cells, and Purkinje named the fluid inside the cell 'protoplasm'.

Applications

- Robert Brown discovered the nucleus in 1831.
- Schleiden and Schwann proposed the cell theory in 1838–39.
- The electron microscope revealed the fine structure of organelles in the twentieth century.

✓ Points to Remember (Exam Focus)

- ✓ Robert Hooke (1665) discovered cells in a thin slice of cork — they were dead cells.
- ✓ Leeuwenhoek was the first to observe live cells (free-living cells in pond water).
- ✓ Cell theory: Schleiden (plants) + Schwann (animals); Virchow added 'omnis cellula e cellula'.
- ✓ Protoplasm — the living substance of the cell — was named by Purkinje.
- ✓ Viruses are an exception to cell theory because they have no cellular structure.

⚠ Common Student Mistakes

- ⚠ Saying Hooke discovered living cells — cork cells were dead and only their walls remained.
- ⚠ Attributing the whole of cell theory to Schleiden and Schwann; Virchow contributed the third part.
- ⚠ Confusing protoplasm (whole living content) with cytoplasm (everything except the nucleus).

**Memory Hook**

'Hooke Hooked the first cell'; and for cell theory, 'Schleiden the plant man, Schwann the animal man, Virchow the ancestor man'.

**Advanced Insights**

Viruses, and the acellular structures of some fungi called coenocytes, are exceptions to strict cell theory. Modern biology therefore states cell theory as applying to all cellular life, with viruses classified separately as non-cellular infectious agents.

4.2 Variety in Cell Number, Shape and Size

🕒 **Estimated time:** 12 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Organisms may be unicellular or multicellular, and cells vary enormously in shape and size — from a bacterium a few micrometres across to an ostrich egg, the largest single cell.

**Real-Life Example**

A single Amoeba does everything a whole human body does, while a human being needs trillions of specialised cells to do the same jobs.

**Detailed Explanation****Definition**

Unicellular organisms consist of a single cell that performs all life functions; multicellular organisms are made of many cells, each group specialised for a particular function.

Key Idea

Cell size is not related to the size of the organism — an elephant and a mouse have cells of similar size; the elephant simply has more of them. Cell shape, however, is closely related to the cell's function.

Working Principle

A cell's shape follows its job: a nerve cell is long and branched to carry messages over distances, a red blood cell is biconcave to maximise surface area for oxygen, a muscle cell is spindle-shaped to contract, and Amoeba has no fixed shape so that it can form pseudopodia.

Applications

- Smallest cell: Mycoplasma (PPLO), about 0.1 micrometre.
- Largest cell: the ostrich egg; the longest cell: the nerve cell.
- Human red blood cell is about 7 micrometres and has no nucleus when mature.

✓ Points to Remember (Exam Focus)

- ✓ Unicellular: Amoeba, Paramecium, bacteria, yeast, Chlamydomonas.
- ✓ Multicellular: all plants and animals, including humans.
- ✓ Cell size varies from 0.1 μm (Mycoplasma) to the ostrich egg, the largest cell.
- ✓ Cell shape depends on function — nerve, muscle, blood and epithelial cells all differ.
- ✓ The number of cells, not their size, determines the size of an organism.

⚠ Common Student Mistakes

- ⚠ Believing bigger animals have bigger cells — they have more cells, not larger ones.
- ⚠ Calling the ostrich egg the largest 'organism cell' without noting that the yolk is the single cell.
- ⚠ Saying Amoeba has a definite shape; it constantly changes shape using pseudopodia.



Memory Hook

'Shape Suits the Service' — a cell's shape always matches the service it performs.



Advanced Insights

Cell size is limited by the surface-area-to-volume ratio. As a cell grows, volume increases as the cube of the radius but surface area only as the square, so beyond a certain size the membrane cannot exchange enough material — which is why organisms grow by adding cells rather than enlarging them.

II Quick Recap — Topics Covered So Far

- **Discovery of the Cell and Cell Theory** — Robert Hooke first saw dead cork cells in 1665, and the cell theory that followed states that all living organisms are made of cells and that every cell comes from a pre-existing cell.
- **Variety in Cell Number, Shape and Size** — Organisms may be unicellular or multicellular, and cells vary enormously in shape and size — from a bacterium a few micrometres across to an ostrich egg, the largest single cell.

4.3 Cell Structure — Membrane, Cytoplasm and Nucleus

Estimated time: 16 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

Every cell has three basic parts: a cell membrane that controls what enters and leaves, cytoplasm where the organelles sit, and a nucleus that controls all cell activity.

Real-Life Example

Think of a cell as a factory — the membrane is the security gate, the cytoplasm is the shop floor with machines, and the nucleus is the manager's office holding the instruction manuals.

Detailed Explanation

Definition

The cell membrane (plasma membrane) is a thin, selectively permeable covering; cytoplasm is the jelly-like substance between membrane and nucleus containing the organelles; the nucleus is the dense spherical body that controls the cell.

Key Idea

The membrane is selectively permeable — it allows only certain substances to pass, which is how a cell keeps its internal environment different from the outside. The nucleus contains chromosomes made of DNA, which carry the instructions for every activity and for inheritance.

Working Principle

The nucleus is bounded by a nuclear membrane with pores, and contains nucleoplasm, a nucleolus and chromatin. During cell division, chromatin condenses into chromosomes. Organelles suspended in the cytoplasm — mitochondria, ribosomes, endoplasmic reticulum, Golgi bodies, lysosomes, plastids — each carry out a specific task.

Applications

- Mitochondria are the 'powerhouses' — they release energy as ATP during respiration.
- Chloroplasts (in plant cells only) trap sunlight for photosynthesis.
- Lysosomes are the 'suicide bags' — they digest worn-out parts and foreign material.

Points to Remember (Exam Focus)

- ✓ The cell membrane is present in ALL cells, plant and animal, and is selectively permeable.
- ✓ The nucleus controls all activities and carries hereditary information in the chromosomes.
- ✓ A cell without a well-defined nucleus is prokaryotic (bacteria, blue-green algae); with a nucleus it is eukaryotic.
- ✓ Mature human red blood cells have no nucleus; sieve tube cells of plants also lack one.
- ✓ Cytoplasm + nucleus together make up the protoplasm.

⚠ Common Student Mistakes

- ⚠ Saying only plant cells have a cell membrane — every cell has one; only plant cells have a cell WALL in addition.
- ⚠ Confusing the nucleolus (a small body inside the nucleus) with the nucleus itself.
- ⚠ Thinking prokaryotes have no genetic material — they have DNA, just not enclosed in a nuclear membrane.



Memory Hook

'Membrane = Gate, Cytoplasm = Factory floor, Nucleus = Manager' — and 'Mito = Motor, Lyso = Litter-bin, Chloro = Cook'.



Advanced Insights

Mitochondria and chloroplasts have their own circular DNA and ribosomes resembling those of bacteria. The endosymbiotic theory explains this: they were once free-living bacteria engulfed by an ancestral cell, which is also why mitochondrial DNA is inherited only from the mother.

4.4 Comparison of Plant and Animal Cells

🕒 **Estimated time:** 12 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

Plant cells differ from animal cells in having a rigid cell wall, plastids and one large central vacuole, while animal cells have centrioles and many small vacuoles.



Real-Life Example

A wilted plant becomes crisp again after watering because water refills its large vacuoles — an animal cell, having no such vacuole, could never do this.

Detailed Explanation

Definition

A plant cell is a eukaryotic cell bounded by a cellulose cell wall outside its membrane; an animal cell is a eukaryotic cell bounded only by a plasma membrane.

Key Idea

Three features are present in plant cells and absent in animal cells: cell wall, plastids and a large central vacuole. One feature — the centriole — is present in animal cells and absent in most plant cells.

Working Principle

The cellulose cell wall gives the plant cell a fixed rectangular shape and protects it from bursting when it takes in water. The large vacuole stores cell sap and provides turgidity, keeping non-woody plants upright.

Applications

- Chloroplasts (green plastids) make the leaf green and carry out photosynthesis.
- Chromoplasts give colour to flowers and fruits; leucoplasts store starch, oil and protein.
- The rigidity of the cell wall is the reason wood and cotton fibre are strong.

✓ Points to Remember (Exam Focus)

- ✓ Present only in plant cells: cell wall, plastids, large central vacuole.
- ✓ Present only in animal cells: centriole (and generally many small vacuoles).
- ✓ Common to both: cell membrane, cytoplasm, nucleus, mitochondria, ribosomes, ER, Golgi bodies.
- ✓ The plant cell wall is made of cellulose; the fungal cell wall is made of chitin.
- ✓ Plant cells are usually larger and have a fixed, regular shape.

⚠ Common Student Mistakes

- ⚠ Saying animal cells have no vacuoles — they have small, temporary ones; only the large central vacuole is plant-specific.
- ⚠ Listing the cell wall as a feature of all cells — bacteria and fungi have walls too, but of different materials, and animal cells have none.
- ⚠ Writing that plant cells have no mitochondria because they photosynthesise; they respire and need mitochondria too.

Memory Hook

'Plants have a WALL, PLastids and a big Pool (vacuole) — animals have none of the three.'

Advanced Insights

Turgor pressure — the outward push of the vacuole against the cell wall — is what holds a herbaceous plant upright without any skeleton. Plasmolysis, the shrinking of protoplasm away from the wall in a strong salt solution, is the reverse of this and is how salt kills weeds.

II Quick Recap — Topics Covered So Far

- Cell Structure — Membrane, Cytoplasm and Nucleus — Every cell has three basic parts: a cell membrane that controls what enters and leaves, cytoplasm where the organelles sit, and a nucleus that controls all cell activity.
- Comparison of Plant and Animal Cells — Plant cells differ from animal cells in having a rigid cell wall, plastids and one large central vacuole, while animal cells have centrioles and many small vacuoles.

Chapter 4 — End of Chapter Revision

All Memory Hooks

- Discovery of the Cell and Cell Theory: 'Hooke Hooked the first cell'; and for cell theory, 'Schleiden the plant man, Schwann the animal man, Virchow the ancestor man'.
- Variety in Cell Number, Shape and Size: 'Shape Suits the Service' — a cell's shape always matches the service it performs.
- Cell Structure — Membrane, Cytoplasm and Nucleus: 'Membrane = Gate, Cytoplasm = Factory floor, Nucleus = Manager' — and 'Mito = Motor, Lyso = Litter-bin, Chloro = Cook'.
- Comparison of Plant and Animal Cells: 'Plants have a WALL, PLastids and a big Pool (vacuole) — animals have none of the three.'

All Common Mistakes

- ⚠ Saying Hooke discovered living cells — cork cells were dead and only their walls remained.
- ⚠ Attributing the whole of cell theory to Schleiden and Schwann; Virchow contributed the third part.
- ⚠ Confusing protoplasm (whole living content) with cytoplasm (everything except the nucleus).
- ⚠ Believing bigger animals have bigger cells — they have more cells, not larger ones.
- ⚠ Calling the ostrich egg the largest 'organism cell' without noting that the yolk is the single cell.
- ⚠ Saying Amoeba has a definite shape; it constantly changes shape using pseudopodia.
- ⚠ Saying only plant cells have a cell membrane — every cell has one; only plant cells have a cell WALL in addition.
- ⚠ Confusing the nucleolus (a small body inside the nucleus) with the nucleus itself.
- ⚠ Thinking prokaryotes have no genetic material — they have DNA, just not enclosed in a nuclear membrane.
- ⚠ Saying animal cells have no vacuoles — they have small, temporary ones; only the large central vacuole is plant-specific.
- ⚠ Listing the cell wall as a feature of all cells — bacteria and fungi have walls too, but of different materials, and animal cells have none.
- ⚠ Writing that plant cells have no mitochondria because they photosynthesise; they respire and need mitochondria too.

Quick Revision Section

- Discovery of the Cell and Cell Theory: Robert Hooke first saw dead cork cells in 1665, and the cell theory that followed states that all living organisms are made of cells and that every cell comes from a pre-existing cell.

- Variety in Cell Number, Shape and Size: Organisms may be unicellular or multicellular, and cells vary enormously in shape and size — from a bacterium a few micrometres across to an ostrich egg, the largest single cell.
- Cell Structure — Membrane, Cytoplasm and Nucleus: Every cell has three basic parts: a cell membrane that controls what enters and leaves, cytoplasm where the organelles sit, and a nucleus that controls all cell activity.
- Comparison of Plant and Animal Cells: Plant cells differ from animal cells in having a rigid cell wall, plastids and one large central vacuole, while animal cells have centrioles and many small vacuoles.

Chapter 5: Reproduction in Animals

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that living organisms produce young ones of their own kind.
- Basic knowledge of the cell and the nucleus.
- Familiarity with the life cycle of a butterfly or frog.
- Understanding that some animals lay eggs and others give birth.

5.1 Sexual Reproduction and Fertilization

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

In sexual reproduction a male gamete (sperm) fuses with a female gamete (ovum) to form a zygote, and this fusion is called fertilization.

Real-Life Example

A hen's egg you eat is an unfertilized ovum; the same egg, if fertilized, would have developed into a chick inside the shell.

Detailed Explanation

Definition

Sexual reproduction is the production of a new individual by the fusion of male and female gametes. Fertilization is the fusion of a sperm with an ovum to form a single cell called the zygote.

Key Idea

Gametes are produced by the reproductive organs — sperms in the testes of the male and ova in the ovaries of the female. Each gamete carries half the chromosomes, so the zygote formed carries a full set with characters from both parents.

Working Principle

The sperm is a tiny motile cell with a head, middle piece and long tail; the ovum is a large, round, non-motile cell. Only one sperm fuses with the ovum. The nuclei of the two gametes fuse, forming the zygote — the very first cell of the new individual.

Applications

- Human male reproductive system: testes, sperm ducts, penis; female: ovaries, oviducts, uterus.
- In-vitro fertilization (IVF) achieves fertilization in a laboratory dish, producing 'test-tube babies'.
- Only one ovum is normally released by a human ovary each month.

✓ Points to Remember (Exam Focus)

- ✓ Zygote = sperm + ovum; it is a single cell and the beginning of a new individual.
- ✓ Testes produce sperms; ovaries produce ova.
- ✓ The sperm is much smaller than the ovum and is the only motile gamete.
- ✓ Variation arises in sexual reproduction because characters come from two parents.
- ✓ The offspring is never exactly identical to either parent.

⚠ Common Student Mistakes

- ⚠ Saying many sperms fertilize one ovum — only one sperm nucleus fuses with the ovum.
- ⚠ Confusing the zygote (single fertilized cell) with the embryo (the developing multicellular stage).
- ⚠ Thinking the ovum is motile; it is carried along the oviduct, it does not swim.

**Memory Hook**

'Sperm + Ovum = Zygote' — 'Two halves make one whole'.

**Advanced Insights**

Once a sperm enters the ovum, the egg membrane changes chemically within seconds — the cortical reaction — so that no second sperm can enter. This 'block to polyspermy' guarantees the zygote receives exactly one set of chromosomes from each parent.

5.2 Internal and External Fertilization

Estimated time: 12 mins **Difficulty:** Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Fertilization inside the female body is internal fertilization; fertilization outside the body, usually in water, is external fertilization.

**Real-Life Example**

A frog lays hundreds of eggs in a pond and the male sheds sperms over them — but a cow carries a single calf inside her for nine months.

**Detailed Explanation****Definition**

Internal fertilization takes place inside the body of the female; external fertilization takes place outside the body of the female, generally in water.

Key Idea

External fertilization is far less certain — gametes may be washed away or eaten — so animals that use it release very large numbers of eggs and sperms. Internal fertilization is highly reliable, so far fewer eggs are needed.

Working Principle

In external fertilization the female lays eggs in water and the male immediately releases sperms over them; the sperms swim to the eggs through the water. In internal fertilization sperms are deposited inside the female body, where they meet the ovum in the oviduct.

Applications

- External fertilization: frogs, fish, starfish — all aquatic.
- Internal fertilization: humans, cows, dogs, hens, birds, reptiles.
- Hens have internal fertilization but lay eggs — internal fertilization does not require live birth.

✓ Points to Remember (Exam Focus)

- ✓ External fertilization needs water as a medium and produces very many eggs.
- ✓ Internal fertilization occurs in the oviduct of the female.
- ✓ Birds and reptiles show internal fertilization but are oviparous (egg-laying).
- ✓ Fewer eggs survive with external fertilization, so survival rate is low.
- ✓ Fertilization always means the fusion of a sperm nucleus with an ovum nucleus.

⚠ Common Student Mistakes

- ⚠ Assuming internal fertilization always means the young are born alive — hens are the standard counter-example.
- ⚠ Thinking external fertilization can happen on land; it needs water for the sperms to swim.
- ⚠ Confusing 'fertilization' with 'development' — fertilization is one event, development takes much longer.

🧠 Memory Hook

'IN the body = Internal; IN the water = External' — and 'External = Enormous numbers of eggs'.

🚀 Advanced Insights

External fertilization requires precise timing, so many fish and amphibians synchronise spawning with the moon, tide or the first monsoon rain — chemical signals called pheromones released into the water keep males and females in step.

II Quick Recap — Topics Covered So Far

- Sexual Reproduction and Fertilization — In sexual reproduction a male gamete (sperm) fuses with a female gamete (ovum) to form a zygote, and this fusion is called fertilization.
- Internal and External Fertilization — Fertilization inside the female body is internal fertilization; fertilization outside the body, usually in water, is external fertilization.

5.3 Development of Embryo and Metamorphosis

🕒 **Estimated time:** 13 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

🔍 Quick Summary

The zygote divides repeatedly to form an embryo, which develops into a foetus, and in some animals the young one changes dramatically into an adult through metamorphosis.

🌐 Real-Life Example

A tadpole with a tail and gills becomes a four-legged, lung-breathing frog — the same animal, completely rebuilt.

📖 Detailed Explanation

Definition

An embryo is the early multicellular stage formed by repeated division of the zygote; a foetus is the later stage in which body parts are identifiable. Metamorphosis is the drastic change of body form from larva to adult.

Key Idea

The zygote divides again and again to form a hollow ball of cells that embeds in the wall of the uterus — this is implantation. Once body parts such as hands, legs, eyes and ears can be recognised, the developing individual is called a foetus.

Working Principle

In humans the gestation period is about nine months, after which the fully developed baby is born. In frogs, the fertilized egg becomes a tadpole that swims with a tail and gills, and thyroid hormone then triggers metamorphosis into the adult frog.

Applications

- Viviparous animals give birth to live young: humans, cows, dogs, cats, whales.
- Oviparous animals lay eggs: hens, frogs, lizards, butterflies, fish.
- Silk moth and butterfly show complete metamorphosis: egg → larva → pupa → adult.

✓ **Points to Remember (Exam Focus)**

- ✓ Zygote → embryo → foetus → young one; implantation happens in the uterus.
- ✓ Human gestation period ≈ 9 months.

- ✓ Viviparous = live young; Oviparous = eggs laid outside.
- ✓ Metamorphosis in frogs is controlled by thyroxine from the thyroid gland.
- ✓ Butterfly life cycle: egg, caterpillar (larva), pupa, adult.

⚠ Common Student Mistakes

- ⚠ Calling a tadpole a 'baby frog' — it is a larva with a completely different body plan.
- ⚠ Saying all egg-laying animals show metamorphosis; a chick hatches looking like a small hen.
- ⚠ Confusing the order embryo → foetus by reversing them.

🧠 Memory Hook

'Zygote, Embryo, Foetus, Baby — ZEFB, the growing-up ladder' — and 'Tadpole needs Thyroxine to Transform'.

🚀 Advanced Insights

If a tadpole is fed thyroxine it metamorphoses early into a miniature frog; if the thyroid is removed it stays a tadpole and simply grows larger. This classic experiment showed for the first time that a single hormone can control an entire developmental programme.

5.4 Asexual Reproduction and Cloning

🕒 **Estimated time:** 13 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

🔍 Quick Summary

Asexual reproduction produces new individuals from a single parent without gametes — by binary fission or budding — and the offspring are genetically identical to the parent.

🌐 Real-Life Example

The tiny buds you can see on a Hydra in pond water grow into complete new Hydra, and Dolly the sheep was an artificial clone made in a laboratory.

📖 Detailed Explanation

Definition

Asexual reproduction is the production of a new individual from a single parent without the fusion of gametes. A clone is an individual genetically identical to its parent.

Key Idea

Because there is only one parent and no fusion of gametes, the offspring receives an exact copy of the parent's genetic material. This gives rapid multiplication but almost no variation.

Working Principle

In binary fission (Amoeba, bacteria) the nucleus divides first, then the cell splits into two equal daughter cells. In budding (Hydra, yeast) a small outgrowth forms on the parent, develops into a miniature individual and then detaches.

Applications

- Binary fission: Amoeba, Paramecium, bacteria, Leishmania.
- Budding: Hydra and yeast.
- Dolly the sheep (1996, Ian Wilmut) was the first mammal cloned from an adult body cell.

✓ Points to Remember (Exam Focus)

- ✓ Asexual reproduction needs only one parent and no gametes.
- ✓ Offspring are exact copies (clones) of the parent — no variation.
- ✓ Binary fission produces two equal cells; budding produces one small outgrowth.
- ✓ Advantage: very rapid multiplication; Disadvantage: no variation, so poor adaptability.
- ✓ Dolly the sheep was created by transferring a nucleus from a body cell into an enucleated egg.

⚠ Common Student Mistakes

- ⚠ Saying budding occurs in Amoeba — Amoeba reproduces by binary fission; Hydra and yeast bud.
- ⚠ Thinking asexual reproduction produces slight variation; it does not, except by mutation.
- ⚠ Believing clones are identical in every way — environment still influences appearance and behaviour.

🧠 Memory Hook

'Fission = Fifty-fifty split; Budding = a Bump that Breaks off' — and 'One parent, One copy'.

🚀 Advanced Insights

Lack of variation is asexual reproduction's fatal weakness. A population of genetically identical individuals can be wiped out by a single new disease — as happened to the Gros Michel banana, which was propagated only by cloning and was destroyed by Panama disease.

II Quick Recap — Topics Covered So Far

- Development of Embryo and Metamorphosis — The zygote divides repeatedly to form an embryo, which develops into a foetus, and in some animals the young one changes dramatically into an adult through metamorphosis.

- **Asexual Reproduction and Cloning** — Asexual reproduction produces new individuals from a single parent without gametes — by binary fission or budding — and the offspring are genetically identical to the parent.

Chapter 5 — End of Chapter Revision

All Memory Hooks

- Sexual Reproduction and Fertilization: 'Sperm + Ovum = Zygote' — 'Two halves make one whole'.
- Internal and External Fertilization: 'IN the body = Internal; IN the water = External' — and 'External = Enormous numbers of eggs'.
- Development of Embryo and Metamorphosis: 'Zygote, Embryo, Foetus, Baby — ZEFB, the growing-up ladder' — and 'Tadpole needs Thyroxine to Transform'.
- Asexual Reproduction and Cloning: 'Fission = Fifty-fifty split; Budding = a Bump that Breaks off' — and 'One parent, One copy'.

All Common Mistakes

-  Saying many sperms fertilize one ovum — only one sperm nucleus fuses with the ovum.
-  Confusing the zygote (single fertilized cell) with the embryo (the developing multicellular stage).
-  Thinking the ovum is motile; it is carried along the oviduct, it does not swim.
-  Assuming internal fertilization always means the young are born alive — hens are the standard counter-example.
-  Thinking external fertilization can happen on land; it needs water for the sperms to swim.
-  Confusing 'fertilization' with 'development' — fertilization is one event, development takes much longer.
-  Calling a tadpole a 'baby frog' — it is a larva with a completely different body plan.
-  Saying all egg-laying animals show metamorphosis; a chick hatches looking like a small hen.
-  Confusing the order embryo → foetus by reversing them.
-  Saying budding occurs in Amoeba — Amoeba reproduces by binary fission; Hydra and yeast bud.
-  Thinking asexual reproduction produces slight variation; it does not, except by mutation.
-  Believing clones are identical in every way — environment still influences appearance and behaviour.

Quick Revision Section

- Sexual Reproduction and Fertilization: In sexual reproduction a male gamete (sperm) fuses with a female gamete (ovum) to form a zygote, and this fusion is called fertilization.
- Internal and External Fertilization: Fertilization inside the female body is internal fertilization; fertilization outside the body, usually in water, is external fertilization.
- Development of Embryo and Metamorphosis: The zygote divides repeatedly to form an embryo, which develops into a foetus, and in some animals the young one changes dramatically into an adult through metamorphosis.

- **Asexual Reproduction and Cloning:** Asexual reproduction produces new individuals from a single parent without gametes — by binary fission or budding — and the offspring are genetically identical to the parent.

Chapter 6: Reaching the Age of Adolescence

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Basic idea of reproduction in animals and the role of testes and ovaries.
- Knowledge that the body grows in height and changes with age.
- Familiarity with the idea of a gland and a secretion.
- Understanding that some characters are inherited from parents.

6.1 Adolescence, Puberty and Changes at Puberty

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Adolescence is the period between about 11 and 19 years of age, during which puberty brings a sudden growth spurt and the development of secondary sexual characters.

Real-Life Example

The sudden change in a boy's voice and the rapid growth in height that classmates notice between classes 7 and 9 are both puberty at work.

Detailed Explanation

Definition

Adolescence is the period of life beginning at about 11 years and ending at about 18 or 19 years, when the body changes from that of a child to that of an adult. Puberty is the stage at which the reproductive organs become functional.

Key Idea

Puberty ends when the reproductive organs become fully functional; adolescence continues a little longer. Adolescents are also called teenagers because this period roughly covers ages 13 to 19.

Working Principle

The most obvious change is the sudden increase in height caused by rapid lengthening of the long bones of the arms and legs. Body shape changes, the voice deepens in boys as the larynx grows, sweat and sebaceous glands become more active causing acne, and the sex organs mature.

Applications

- Adam's apple — the enlarged larynx visible in boys.
- Estimating full adult height from the percentage of height already attained.
- Understanding why acne appears at this age and needs cleanliness rather than worry.

✓ Points to Remember (Exam Focus)

- ✓ Adolescence: about 11 to 19 years; puberty is the part of it when reproductive organs mature.
- ✓ Growth spurt: the fastest increase in height in the whole of life after infancy.
- ✓ In boys the larynx (voice box) grows large, producing the Adam's apple and a deep voice.
- ✓ Increased activity of sweat and sebaceous glands causes acne and pimples.
- ✓ Secondary sexual characters: facial hair and broad shoulders in boys; breast development and wider hips in girls.

⚠ Common Student Mistakes

- ⚠ Using 'adolescence' and 'puberty' as the same word — puberty ends earlier, when reproductive organs become functional.
- ⚠ Thinking girls and boys start puberty at the same age; girls usually begin about a year or two earlier.
- ⚠ Believing acne is caused by diet alone; the main cause is increased secretion of the sebaceous glands.

**Memory Hook**

'Puberty = the Power switch of the reproductive organs; Adolescence = the whole Adjustment period around it.'

**Advanced Insights**

The growth spurt ends when the epiphyseal plates — the growth zones at the ends of long bones — fuse under the influence of sex hormones. Once fused, no further increase in height is possible, which is why height stops changing after adolescence.

6.2 Hormones and the Endocrine System

🕒 Estimated time: 16 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Endocrine glands release hormones directly into the bloodstream, and these chemical messengers control growth, metabolism and the changes of puberty.

**Real-Life Example**

The pounding heart and rush of energy you feel just before an exam or after a near-miss on the road is adrenaline released by the adrenal glands.

Detailed Explanation

Definition

A hormone is a chemical messenger secreted in very small amounts by an endocrine gland and carried by the blood to a target organ. Endocrine (ductless) glands pour secretions straight into the blood; exocrine glands use ducts.

Key Idea

The pituitary gland, controlled by the hypothalamus, is the master gland: it secretes growth hormone and also the hormones that switch on the testes, ovaries and thyroid. Sex hormones from the testes and ovaries are what actually produce the secondary sexual characters.

Working Principle

The pituitary releases FSH and LH into the blood; these reach the testes and ovaries and stimulate them to produce testosterone and oestrogen respectively. These sex hormones then act on the body to produce the changes of puberty.

Applications

- Thyroid → thyroxine; its deficiency causes goitre, and iodised salt prevents it.
- Pancreas → insulin, which controls blood sugar; its deficiency causes diabetes.
- Adrenal glands → adrenaline, the 'emergency' or fight-or-flight hormone.

✓ Points to Remember (Exam Focus)

- ✓ Pituitary is the master gland; it is controlled by the hypothalamus.
- ✓ Testes → testosterone (male hormone); Ovaries → oestrogen (female hormone).
- ✓ Thyroid → thyroxine (needs iodine); deficiency → goitre.
- ✓ Pancreas → insulin; deficiency → diabetes.
- ✓ Adrenal glands → adrenaline, released during fear, anger or excitement.

⚠ Common Student Mistakes

- ⚠ Calling the pancreas purely an endocrine gland — it is both endocrine (insulin) and exocrine (digestive juice).
- ⚠ Saying the hypothalamus is the master gland; the pituitary is, though the hypothalamus controls it.
- ⚠ Confusing thyroxine deficiency (goitre) with insulin deficiency (diabetes).

Memory Hook

'Pituitary is the Principal, all other glands are the teachers' — and 'Adrenaline for Action, Insulin for Sugar, Thyroxine needs Iodine'.

Advanced Insights

Hormone levels are held steady by negative feedback: rising thyroxine in the blood suppresses the pituitary hormone that stimulated the thyroid in the first place. This self-correcting loop is the same principle a thermostat uses to hold a room at a set temperature.

II Quick Recap — Topics Covered So Far

- **Adolescence, Puberty and Changes at Puberty** — Adolescence is the period between about 11 and 19 years of age, during which puberty brings a sudden growth spurt and the development of secondary sexual characters.
- **Hormones and the Endocrine System** — Endocrine glands release hormones directly into the bloodstream, and these chemical messengers control growth, metabolism and the changes of puberty.

6.3 Reproductive Phase of Life and the Menstrual Cycle

 **Estimated time:** 13 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

The reproductive phase begins at puberty; in females it is marked by a roughly 28-day menstrual cycle that begins with menarche and ends at menopause.

Real-Life Example

The monthly cycle is the body preparing for a possible pregnancy each month and clearing away the preparation when it does not occur.

Detailed Explanation

Definition

Menarche is the first menstruation, at about 10–12 years; menopause is the permanent stopping of menstruation, at about 45–50 years. The menstrual cycle is the roughly 28-day cycle of changes in the ovary and uterus.

Key Idea

One ovum matures and is released by an ovary about every 28 days — this is ovulation. Meanwhile the uterus lining thickens to receive a possible embryo. If fertilization does not occur, the lining breaks down and is shed as menstrual flow.

Working Principle

The cycle has four phases: menstrual (days 1–5, shedding of the lining), follicular/proliferative (days 6–13, lining rebuilds), ovulation (about day 14, ovum released), and luteal/secretory (days 15–28, lining maintained awaiting implantation).

Applications

- Ovulation on about day 14 is why the middle of the cycle is the fertile period.
- Menstrual hygiene and adequate iron in the diet are important during this phase.

- The reproductive phase in males begins at puberty and continues much longer.

✓ Points to Remember (Exam Focus)

- ✓ Menstrual cycle length $\approx$ 28 days; ovulation occurs around day 14.
- ✓ Menarche = first menstruation; Menopause = end of menstruation (about 45–50 years).
- ✓ Menstruation lasts about 3–5 days and is the shedding of the thickened uterine lining.
- ✓ Only one ovum is normally released per cycle, by one of the two ovaries.
- ✓ If the ovum is fertilized, menstruation stops and pregnancy begins.

⚠ Common Student Mistakes

- ⚠ Thinking the ovum is released on day 1 — it is released about mid-cycle, around day 14.
- ⚠ Confusing menarche (the first period) with menopause (the last).
- ⚠ Assuming the cycle is exactly 28 days for everyone; it commonly varies from 21 to 35 days.



Memory Hook

'MenARCHE = the ARCHway in (first); MenoPAUSE = the PAUSE for good (last)' — and 'Day 14, the ovum leaves'.



Advanced Insights

Ovulation is triggered by a sharp surge of luteinising hormone (LH) from the pituitary. Home ovulation-prediction kits work by detecting exactly this LH surge in urine, roughly 24–36 hours before the ovum is released.

6.4 Sex Determination and Reproductive Health

🕒 **Estimated time:** 12 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

The sex of a baby is decided by the sex chromosome carried by the father's sperm, and staying healthy through adolescence needs a balanced diet, hygiene, exercise and avoidance of drugs.



Real-Life Example

Blaming a mother for the sex of her child is biologically wrong — it is always the father's sperm that decides.

Detailed Explanation

Definition

Sex chromosomes are the pair of chromosomes that determine the sex of an individual: XX in a female and XY in a male. Reproductive health is the state of complete physical and mental well-being of the reproductive system.

Key Idea

Every ovum carries an X chromosome. A sperm may carry either X or Y. If an X-bearing sperm fertilizes the ovum the child is a girl (XX); if a Y-bearing sperm does, the child is a boy (XY). The chance is therefore about 50:50 each time.

Working Principle

Humans have 23 pairs of chromosomes, of which 22 pairs are autosomes and one pair is the sex chromosomes. During gamete formation each gamete receives one chromosome of each pair, so the father's sperm decides the sex.

Applications

- A balanced diet rich in protein, iron, calcium and vitamins supports the growth spurt.
- Personal hygiene and daily bathing prevent infections and control body odour and acne.
- AIDS is caused by HIV and spreads by unsafe blood transfusion, infected needles and from mother to child — not by casual contact.

✓ Points to Remember (Exam Focus)

- ✓ Female = XX, Male = XY; the ovum always carries X.
- ✓ The father's sperm (X or Y) determines the sex of the child.
- ✓ Humans have 23 pairs of chromosomes: 22 pairs autosomes + 1 pair sex chromosomes.
- ✓ Adolescents need extra iron (jaggery, green leafy vegetables) and protein.
- ✓ Say no to drugs — they are addictive, harm the body permanently and can spread HIV through shared needles.

⚠ Common Student Mistakes

- ⚠ Saying the mother determines the sex of the baby — she can only contribute an X.
- ⚠ Believing AIDS spreads by touching, sharing food or sitting beside an infected person; it does not.
- ⚠ Thinking junk food is fine during the growth spurt; adolescence has the highest nutritional demand of life.

Memory Hook

'Mother always gives X, Father gives the choice' — XX girl, XY boy.

Advanced Insights

The Y chromosome carries a single decisive gene, SRY, which switches the embryonic gonad into a testis. Everything else about male development follows from the hormones that testis then produces — one gene setting off an entire cascade.

II Quick Recap — Topics Covered So Far

- Reproductive Phase of Life and the Menstrual Cycle — The reproductive phase begins at puberty; in females it is marked by a roughly 28-day menstrual cycle that begins with menarche and ends at menopause.
- Sex Determination and Reproductive Health — The sex of a baby is decided by the sex chromosome carried by the father's sperm, and staying healthy through adolescence needs a balanced diet, hygiene, exercise and avoidance of drugs.

Chapter 6 — End of Chapter Revision

All Memory Hooks

- Adolescence, Puberty and Changes at Puberty: 'Puberty = the Power switch of the reproductive organs; Adolescence = the whole Adjustment period around it.'
- Hormones and the Endocrine System: 'Pituitary is the Principal, all other glands are the teachers' — and 'Adrenaline for Action, Insulin for Sugar, Thyroxine needs Iodine'.
- Reproductive Phase of Life and the Menstrual Cycle: 'MenARCHE = the ARCHway in (first); MenoPAUSE = the PAUSE for good (last)' — and 'Day 14, the ovum leaves'.
- Sex Determination and Reproductive Health: 'Mother always gives X, Father gives the choice' — XX girl, XY boy.

All Common Mistakes

- ⚠ Using 'adolescence' and 'puberty' as the same word — puberty ends earlier, when reproductive organs become functional.
- ⚠ Thinking girls and boys start puberty at the same age; girls usually begin about a year or two earlier.
- ⚠ Believing acne is caused by diet alone; the main cause is increased secretion of the sebaceous glands.
- ⚠ Calling the pancreas purely an endocrine gland — it is both endocrine (insulin) and exocrine (digestive juice).
- ⚠ Saying the hypothalamus is the master gland; the pituitary is, though the hypothalamus controls it.
- ⚠ Confusing thyroxine deficiency (goitre) with insulin deficiency (diabetes).
- ⚠ Thinking the ovum is released on day 1 — it is released about mid-cycle, around day 14.
- ⚠ Confusing menarche (the first period) with menopause (the last).
- ⚠ Assuming the cycle is exactly 28 days for everyone; it commonly varies from 21 to 35 days.
- ⚠ Saying the mother determines the sex of the baby — she can only contribute an X.
- ⚠ Believing AIDS spreads by touching, sharing food or sitting beside an infected person; it does not.
- ⚠ Thinking junk food is fine during the growth spurt; adolescence has the highest nutritional demand of life.

Quick Revision Section

- Adolescence, Puberty and Changes at Puberty: Adolescence is the period between about 11 and 19 years of age, during which puberty brings a sudden growth spurt and the development of secondary sexual characters.
- Hormones and the Endocrine System: Endocrine glands release hormones directly into the bloodstream, and these chemical messengers control growth, metabolism and the changes of puberty.

- Reproductive Phase of Life and the Menstrual Cycle: The reproductive phase begins at puberty; in females it is marked by a roughly 28-day menstrual cycle that begins with menarche and ends at menopause.
- Sex Determination and Reproductive Health: The sex of a baby is decided by the sex chromosome carried by the father's sperm, and staying healthy through adolescence needs a balanced diet, hygiene, exercise and avoidance of drugs.

Chapter 7: Pollution of Air and Water

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge that air is a mixture of gases and that we breathe oxygen.
- Idea of the water cycle — evaporation, condensation and rainfall.
- Awareness that vehicles and factories release smoke.
- Basic understanding that drinking dirty water causes illness.

7.1 Air Pollution and Its Effects

 **Estimated time:** 14 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Air pollution is the contamination of air by harmful substances, mainly from vehicles, industries and the burning of fuels, and it damages health, buildings and the climate.

Real-Life Example

The winter smog over Delhi that makes eyes water, and the yellowing of the Taj Mahal's marble, are two visible faces of air pollution.

Detailed Explanation

Definition

Air pollution is the presence of substances in air in concentrations high enough to harm living organisms or damage materials. Such substances are called air pollutants.

Key Idea

Pollutants come mostly from burning fossil fuels. Sulphur dioxide and nitrogen dioxide dissolve in rain water to form acid rain; carbon monoxide binds to haemoglobin and reduces oxygen carriage; suspended particulate matter causes respiratory disease.

Working Principle

Chlorofluorocarbons (CFCs) released from refrigerators, air conditioners and aerosol sprays rise to the stratosphere and destroy ozone, thinning the ozone layer that shields us from ultraviolet radiation.

Applications

- The Taj Mahal's marble is being corroded and yellowed by acid rain and suspended particles — 'marble cancer'.
- CNG buses were introduced in Delhi to cut vehicular pollution.
- Catalytic converters in cars convert harmful exhaust gases into less harmful ones.

✓ Points to Remember (Exam Focus)

- ✓ Major pollutants: SO₂, NO₂, CO, CFCs, suspended particulate matter (SPM), smoke.
- ✓ Acid rain is caused mainly by sulphur dioxide and nitrogen dioxide dissolving in rainwater.
- ✓ CFCs damage the ozone layer, which protects earth from harmful ultraviolet rays.
- ✓ Carbon monoxide is dangerous because it reduces the oxygen-carrying capacity of blood.
- ✓ The Taj Mahal is damaged by acid rain — a phenomenon called marble cancer.

⚠ Common Student Mistakes

- ⚠ Confusing the ozone layer problem (CFCs) with the greenhouse effect (CO₂) — they are different problems.
- ⚠ Calling carbon dioxide a 'poisonous' gas; it is not poisonous but is a greenhouse gas.
- ⚠ Thinking smog is just fog; smog is smoke plus fog and contains pollutants.



Memory Hook

'CFC Cuts the Cover (ozone); CO₂ Covers the earth like a blanket (warming)' — two different C's, two different problems.



Advanced Insights

The Montreal Protocol of 1987 phased out CFCs worldwide and is the most successful environmental treaty ever signed — the ozone hole over Antarctica is now measurably shrinking and is expected to close by about 2065.

7.2 Greenhouse Effect and Global Warming

🕒 **Estimated time:** 12 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

Greenhouse gases trap the earth's outgoing heat and keep the planet warm; increasing their concentration is raising the average temperature, which is called global warming.



Real-Life Example

A closed car parked in the sun becomes far hotter inside than outside — sunlight enters through the glass but the heat cannot get out. The atmosphere does the same thing on a planetary scale.



Detailed Explanation

Definition

The greenhouse effect is the trapping of the sun's heat by certain gases in the atmosphere. Global warming is the resulting gradual increase in the average temperature of the earth.

Key Idea

The natural greenhouse effect is essential — without it the earth's average temperature would be about -18°C instead of about 15°C . The problem is the enhanced greenhouse effect caused by human emissions.

Working Principle

Short-wavelength sunlight passes through the atmosphere and warms the earth's surface. The surface re-radiates this energy as longer-wavelength infrared, which greenhouse gases absorb and re-emit back downward, trapping heat near the surface.

Applications

- Melting of polar ice caps and glaciers, raising sea level and threatening coastal cities.
- Changed rainfall patterns, more frequent droughts and floods.
- Switching to solar, wind and hydro power reduces CO_2 emissions.

✓ Points to Remember (Exam Focus)

- ✓ Greenhouse gases: carbon dioxide, methane, nitrous oxide, water vapour, CFCs.
- ✓ CO_2 is the main greenhouse gas released by human activity, mostly from burning fossil fuels.
- ✓ Global warming causes melting of glaciers, rise in sea level and changes in rainfall.
- ✓ Deforestation increases CO_2 because fewer trees absorb it.
- ✓ Using public transport, planting trees and using clean fuels reduce the problem.

⚠ Common Student Mistakes

- ⚠ Saying the greenhouse effect is entirely harmful — the natural greenhouse effect keeps earth habitable.
- ⚠ Confusing global warming (temperature rise from CO_2) with ozone depletion (UV from CFCs).
- ⚠ Believing plants alone can absorb all human CO_2 emissions; the rate of emission far exceeds absorption.

🧠 Memory Hook

'Sunlight gets IN, heat can't get OUT' — that is a greenhouse, a parked car, and our atmosphere.

🚀 Advanced Insights

Methane traps roughly 25 times more heat per molecule than CO_2 but survives only about a decade in the atmosphere, whereas CO_2 persists for centuries. Cutting methane therefore slows warming quickly, while cutting CO_2 is what matters over the long run.

II Quick Recap — Topics Covered So Far

- Air Pollution and Its Effects — Air pollution is the contamination of air by harmful substances, mainly from vehicles, industries and the burning of fuels, and it damages health, buildings and the climate.
- Greenhouse Effect and Global Warming — Greenhouse gases trap the earth's outgoing heat and keep the planet warm; increasing their concentration is raising the average temperature, which is called global warming.

7.3 Water Pollution and Potable Water

🕒 **Estimated time:** 14 mins **Difficulty:** ● Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Water pollution is the addition of harmful substances to water bodies from sewage, industrial waste and agricultural runoff; water fit for drinking is called potable water.

Real-Life Example

The froth on the Yamuna and the fish deaths reported after a factory discharge are water pollution; the boiling or filtering of water at home is the household answer to it.

Detailed Explanation

Definition

Water pollution is the contamination of water by substances that make it unfit for use by living organisms. Potable water is water that is safe to drink — free from harmful microorganisms and chemicals.

Key Idea

The three main sources are sewage (domestic waste), industrial effluents containing arsenic, lead and mercury, and agricultural runoff carrying fertilizers and pesticides. Hot water discharged from power plants causes thermal pollution, which lowers dissolved oxygen and kills fish.

Working Principle

Water is purified at home by filtration (removes suspended solids), boiling (kills microorganisms), and chlorination (a chlorine tablet kills germs). Sedimentation, filtration and chlorination in that order are used in municipal treatment plants.

Applications

- The Ganga Action Plan was launched in 1985 to reduce pollution of the river Ganga.
- Chlorine tablets are used to disinfect drinking water during floods.
- Water-borne diseases: cholera, typhoid, dysentery, hepatitis, jaundice.

✓ Points to Remember (Exam Focus)

- ✓ Sources of water pollution: sewage, industrial effluents, agricultural chemicals, hot water (thermal).
- ✓ Potable water is free of germs and harmful chemicals and is safe to drink.
- ✓ Boiling is the simplest and most reliable household method of purification.
- ✓ Excess fertilizer causes eutrophication — algal bloom followed by oxygen depletion.
- ✓ The Ganga Action Plan (1985) was launched to clean the river Ganga.

⚠ Common Student Mistakes

- ⚠ Assuming clear water is safe water; harmful microbes and dissolved chemicals are invisible.
- ⚠ Thinking filtration kills germs — it only removes suspended solids; boiling or chlorination kills germs.
- ⚠ Ignoring thermal pollution as a form of water pollution just because nothing is 'added'.



Memory Hook

'Filter it, Boil it, Chlorinate it' — and 'Clear ≠ Clean'.



Advanced Insights

Eutrophication is a chain reaction: nutrients → algal bloom → algae die → decomposers multiply → dissolved oxygen falls → fish suffocate. It shows that a pollutant need not be poisonous; an excess of a nutrient can be just as lethal.

II Quick Recap — Topics Covered So Far

- Water Pollution and Potable Water — Water pollution is the addition of harmful substances to water bodies from sewage, industrial waste and agricultural runoff; water fit for drinking is called potable water.

Chapter 7 — End of Chapter Revision

All Memory Hooks

- Air Pollution and Its Effects: 'CFC Cuts the Cover (ozone); CO₂ Covers the earth like a blanket (warming)' — two different C's, two different problems.
- Greenhouse Effect and Global Warming: 'Sunlight gets IN, heat can't get OUT' — that is a greenhouse, a parked car, and our atmosphere.
- Water Pollution and Potable Water: 'Filter it, Boil it, Chlorinate it' — and 'Clear ≠ Clean'.

All Common Mistakes

- ⚠ Confusing the ozone layer problem (CFCs) with the greenhouse effect (CO₂) — they are different problems.
- ⚠ Calling carbon dioxide a 'poisonous' gas; it is not poisonous but is a greenhouse gas.
- ⚠ Thinking smog is just fog; smog is smoke plus fog and contains pollutants.
- ⚠ Saying the greenhouse effect is entirely harmful — the natural greenhouse effect keeps earth habitable.
- ⚠ Confusing global warming (temperature rise from CO₂) with ozone depletion (UV from CFCs).
- ⚠ Believing plants alone can absorb all human CO₂ emissions; the rate of emission far exceeds absorption.
- ⚠ Assuming clear water is safe water; harmful microbes and dissolved chemicals are invisible.
- ⚠ Thinking filtration kills germs — it only removes suspended solids; boiling or chlorination kills germs.
- ⚠ Ignoring thermal pollution as a form of water pollution just because nothing is 'added'.

Quick Revision Section

- Air Pollution and Its Effects: Air pollution is the contamination of air by harmful substances, mainly from vehicles, industries and the burning of fuels, and it damages health, buildings and the climate.
- Greenhouse Effect and Global Warming: Greenhouse gases trap the earth's outgoing heat and keep the planet warm; increasing their concentration is raising the average temperature, which is called global warming.
- Water Pollution and Potable Water: Water pollution is the addition of harmful substances to water bodies from sewage, industrial waste and agricultural runoff; water fit for drinking is called potable water.

Chapter 8: Life Processes

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that living things need food, air and water.
- Basic knowledge of the cell and its parts.
- Familiarity with the words photosynthesis and respiration.
- Awareness that blood flows through the body.

8.1 Nutrition in Plants and Animals

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Green plants are autotrophs that make their own food by photosynthesis, while animals are heterotrophs that depend directly or indirectly on plants.

Real-Life Example

Every meal you eat traces back to a green leaf — the wheat in a chapati, and the grass eaten by the cow whose milk you drink.

Detailed Explanation

Definition

Nutrition is the process of taking in food and using it for energy, growth and repair. Autotrophic nutrition is the making of one's own food from simple inorganic substances; heterotrophic nutrition is obtaining ready-made organic food.

Key Idea

Photosynthesis converts carbon dioxide and water into glucose and oxygen using sunlight trapped by chlorophyll. It is the only process on earth that puts energy into the living world, and it releases the oxygen every animal breathes.

Working Principle

$6\text{CO}_2 + 6\text{H}_2\text{O}$, in the presence of sunlight and chlorophyll, gives $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Carbon dioxide enters through stomata, water is drawn up by the roots, and glucose is stored as starch. Heterotrophs are further grouped as holozoic (ingest solid food), saprophytic (feed on dead matter) and parasitic.

Applications

- Iodine test: starch turns blue-black, proving photosynthesis has occurred in a leaf.
- Insectivorous plants such as pitcher plant and Venus flytrap get nitrogen by trapping insects.
- Fungi such as bread mould are saprophytes; *Cuscuta* (amarbel) is a parasitic plant.

✓ Points to Remember (Exam Focus)

- ✓ Photosynthesis occurs in the chloroplasts, mainly in the leaf, and requires chlorophyll.
- ✓ Raw materials: carbon dioxide (from air through stomata) and water (from soil through roots).
- ✓ Products: glucose (stored as starch) and oxygen (released).
- ✓ Autotrophs = green plants; Heterotrophs = all animals and fungi.
- ✓ Heterotrophic types: holozoic (humans, Amoeba), saprophytic (fungi), parasitic (Cuscuta, tapeworm).

⚠ Common Student Mistakes

- ⚠ Saying plants respire only at night — plants respire all the time; photosynthesis only happens in light.
- ⚠ Thinking sunlight is a raw material; sunlight is the energy source, CO₂ and water are the raw materials.
- ⚠ Calling insectivorous plants heterotrophs — they photosynthesise and are autotrophs that supplement nitrogen.



Memory Hook

'AUTO = self-service; HETERO = ordering from outside' — and ' $\text{CO}_2 + \text{H}_2\text{O} + \text{Sun} + \text{Chlorophyll} \rightarrow \text{Sugar} + \text{O}_2$ '.



Advanced Insights

Photosynthesis has two stages: light reactions in the thylakoid membranes split water and produce ATP and NADPH, and the light-independent Calvin cycle in the stroma uses these to fix CO₂ into sugar. This is why the two are often called the light and dark reactions.

8.2 Digestion, Respiration and Transportation in Humans

🕒 **Estimated time:** 16 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

The digestive system breaks food into absorbable form, respiration releases energy from that food, and the circulatory system carries both nutrients and oxygen to every cell.



Real-Life Example

The rumbling of an empty stomach, the faster breathing after running up stairs, and the pulse you feel at your wrist are these three systems announcing themselves.



Detailed Explanation

Definition

Digestion is the breakdown of complex food into simple absorbable substances. Respiration is the release of energy from food. Transportation is the movement of substances around the body by the blood.

Key Idea

Digestion begins in the mouth with salivary amylase acting on starch, continues in the stomach where pepsin and hydrochloric acid act on protein, and is completed in the small intestine by bile, pancreatic juice and intestinal juice. Absorption occurs through the villi of the small intestine.

Working Principle

Aerobic respiration: $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy (ATP)}$. The blood, pumped by the four-chambered heart, carries oxygen bound to haemoglobin in red blood cells, and returns carbon dioxide to the lungs.

Applications

- Anaerobic respiration in muscles during heavy exercise produces lactic acid and causes cramps.
- Yeast respire anaerobically to give alcohol and carbon dioxide — used in brewing and baking.
- Blood pressure is normally about 120/80 mm of mercury in a healthy adult.

✓ Points to Remember (Exam Focus)

- ✓ Alimentary canal: mouth → oesophagus → stomach → small intestine → large intestine → anus.
- ✓ Largest gland in the human body is the liver; it secretes bile which emulsifies fat.
- ✓ Absorption of digested food occurs through the villi of the small intestine.
- ✓ Aerobic respiration needs oxygen and releases much more energy than anaerobic respiration.
- ✓ The human heart has four chambers: two atria and two ventricles.

⚠ Common Student Mistakes

- ⚠ Saying digestion is completed in the stomach — most digestion and all absorption happen in the small intestine.
- ⚠ Confusing breathing (exchange of gases) with respiration (release of energy inside cells).
- ⚠ Writing that veins always carry deoxygenated blood; the pulmonary vein carries oxygenated blood.

**Memory Hook**

'Mouth, Oesophagus, Stomach, Small, Large — MOSSL, the food's journey' — and 'Breathing is the doorway, respiration is the furnace'.

**Advanced Insights**

The small intestine is about 6 metres long and its villi and microvilli raise the absorptive surface to roughly 250 square metres — about the area of a tennis court packed into the abdomen. Surface area is the recurring design solution in every exchange organ.

II Quick Recap — Topics Covered So Far

- Nutrition in Plants and Animals — Green plants are autotrophs that make their own food by photosynthesis, while animals are heterotrophs that depend directly or indirectly on plants.
- Digestion, Respiration and Transportation in Humans — The digestive system breaks food into absorbable form, respiration releases energy from that food, and the circulatory system carries both nutrients and oxygen to every cell.

8.3 Excretion and Control of the Internal Environment

 **Estimated time:** 12 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

Excretion removes the nitrogenous wastes produced by the body's chemical reactions, and in humans it is carried out mainly by the kidneys, which filter the blood.

Real-Life Example

A dialysis machine does mechanically what a healthy kidney does continuously — filtering waste from the blood of a patient whose kidneys have failed.

Detailed Explanation

Definition

Excretion is the removal of harmful metabolic wastes, especially nitrogenous wastes such as urea, from the body. The excretory system in humans consists of two kidneys, two ureters, a urinary bladder and the urethra.

Key Idea

Each kidney contains about a million tiny filtering units called nephrons. Blood is filtered under pressure, and then useful substances — glucose, amino acids, most water and salts — are reabsorbed, leaving urine.

Working Principle

Urea is formed in the liver from the breakdown of excess amino acids, carried by the blood to the kidneys, filtered out and excreted in urine. Skin excretes water and salts as sweat, and lungs excrete carbon dioxide and water vapour.

Applications

- Dialysis is used when the kidneys fail, to remove urea artificially from the blood.
- Plants excrete by shedding leaves, storing waste in vacuoles, and releasing gums and resins.
- Sweating cools the body as well as removing salts and urea.

✓ Points to Remember (Exam Focus)

- ✓ Excretory organs: kidneys (urea), lungs (CO₂), skin (sweat), liver (bile pigments).
- ✓ The functional unit of the kidney is the nephron.
- ✓ Urea is produced in the liver, not in the kidney.
- ✓ Urine is about 95% water, with urea, uric acid and salts.
- ✓ Plants have no specialised excretory system.

⚠ Common Student Mistakes

- ⚠ Saying urea is made in the kidney — it is made in the liver and only filtered by the kidney.
- ⚠ Confusing excretion (removal of metabolic waste) with egestion (removal of undigested food as faeces).
- ⚠ Forgetting that lungs and skin are also excretory organs.



Memory Hook

'Liver Makes it, Kidney Takes it' — urea is made in the liver and removed by the kidney.



Advanced Insights

The kidneys filter about 180 litres of fluid a day but produce only about 1.5 litres of urine — over 99% is reabsorbed. This enormous filter-then-reclaim strategy lets the body finely adjust water and salt balance moment by moment.

II Quick Recap — Topics Covered So Far

- Excretion and Control of the Internal Environment — Excretion removes the nitrogenous wastes produced by the body's chemical reactions, and in humans it is carried out mainly by the kidneys, which filter the blood.

Chapter 8 — End of Chapter Revision

All Memory Hooks

- Nutrition in Plants and Animals: 'AUTO = self-service; HETERO = ordering from outside' — and ' $\text{CO}_2 + \text{H}_2\text{O} + \text{Sun} + \text{Chlorophyll} \rightarrow \text{Sugar} + \text{O}_2$ '.
- Digestion, Respiration and Transportation in Humans: 'Mouth, Oesophagus, Stomach, Small, Large — MOSSL, the food's journey' — and 'Breathing is the doorway, respiration is the furnace'.
- Excretion and Control of the Internal Environment: 'Liver Makes it, Kidney Takes it' — urea is made in the liver and removed by the kidney.

All Common Mistakes

-  Saying plants respire only at night — plants respire all the time; photosynthesis only happens in light.
-  Thinking sunlight is a raw material; sunlight is the energy source, CO_2 and water are the raw materials.
-  Calling insectivorous plants heterotrophs — they photosynthesise and are autotrophs that supplement nitrogen.
-  Saying digestion is completed in the stomach — most digestion and all absorption happen in the small intestine.
-  Confusing breathing (exchange of gases) with respiration (release of energy inside cells).
-  Writing that veins always carry deoxygenated blood; the pulmonary vein carries oxygenated blood.
-  Saying urea is made in the kidney — it is made in the liver and only filtered by the kidney.
-  Confusing excretion (removal of metabolic waste) with egestion (removal of undigested food as faeces).
-  Forgetting that lungs and skin are also excretory organs.

Quick Revision Section

- Nutrition in Plants and Animals: Green plants are autotrophs that make their own food by photosynthesis, while animals are heterotrophs that depend directly or indirectly on plants.
- Digestion, Respiration and Transportation in Humans: The digestive system breaks food into absorbable form, respiration releases energy from that food, and the circulatory system carries both nutrients and oxygen to every cell.
- Excretion and Control of the Internal Environment: Excretion removes the nitrogenous wastes produced by the body's chemical reactions, and in humans it is carried out mainly by the kidneys, which filter the blood.

Chapter 9: Tissues

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge that the cell is the basic unit of life.
- Difference between a plant cell and an animal cell.
- Idea that organs such as the heart and leaf do specific jobs.
- Familiarity with the terms unicellular and multicellular.

9.1 Plant Tissues — Meristematic and Permanent

 **Estimated time:** 15 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Plant tissues are either meristematic, which divide and cause growth, or permanent, which have lost the power to divide and perform a fixed function.

Real-Life Example

A plant grows taller at its tips and thicker at its stem because different meristems are working in different places — which is also why pruning a shoot tip makes a hedge grow bushy.

Detailed Explanation

Definition

A tissue is a group of similar cells performing a specific function. Meristematic tissue consists of actively dividing cells; permanent tissue consists of cells that have stopped dividing and have taken a definite shape and function.

Key Idea

Growth in plants is restricted to certain regions where meristems sit. Apical meristem at the tips increases length, intercalary meristem at the base of leaves and internodes also adds length, and lateral meristem (cambium) increases girth.

Working Principle

Simple permanent tissues are made of one cell type — parenchyma (packing and storage), collenchyma (flexible support in stems and leaf stalks) and sclerenchyma (hard, dead cells giving rigidity). Complex permanent tissues are made of more than one cell type — xylem and phloem.

Applications

- Chlorenchyma is parenchyma with chloroplasts and performs photosynthesis.
- Aerenchyma has large air cavities and helps aquatic plants float.
- Coconut husk fibre and jute fibre are sclerenchyma.

✓ Points to Remember (Exam Focus)

- ✓ Apical meristem → increase in length; Lateral meristem (cambium) → increase in girth.
- ✓ Parenchyma: living, thin-walled, stores food; Collenchyma: living, flexible support; Sclerenchyma: dead, thick lignified walls, rigidity.
- ✓ Xylem carries water and minerals upwards only — it is unidirectional.
- ✓ Phloem carries food from leaves to all parts — it is bidirectional.
- ✓ Xylem is made of tracheids, vessels, xylem parenchyma and xylem fibres.

⚠ Common Student Mistakes

- ⚠ Saying phloem carries water — xylem carries water; phloem carries food.
- ⚠ Calling sclerenchyma a living tissue; its cells are dead at maturity.
- ⚠ Confusing apical meristem (length) with lateral meristem (girth).



Memory Hook

'Xylem = eXit upward with water; Phloem = Food, both ways' — and 'Para stores, Colla bends, Sclera hardens'.



Advanced Insights

Xylem transport works by transpiration pull: water evaporating from the leaves creates tension that pulls a continuous column of water up the vessels, held together by cohesion. No pump is involved — the plant is drinking through a straw hundreds of times its own width.

9.2 Animal Tissues — Epithelial and Connective

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★



Quick Summary

Epithelial tissue covers and lines body surfaces, while connective tissue binds, supports and transports — blood and bone are both connective tissues.



Real-Life Example

Your skin is epithelium sealing the outside world away, and the blood beneath it is a connective tissue delivering supplies to every cell.

📖 Detailed Explanation

Definition

Epithelial tissue is the covering or protective tissue that lines the outer surface of the body and the inner surface of organs and cavities. Connective tissue connects, supports and binds different tissues and organs, and has an abundant matrix.

Key Idea

Epithelium always rests on a basement membrane and has very little intercellular material; connective tissue is the opposite — the cells are few and widely separated in a large matrix, which may be fluid (blood), solid (bone) or jelly-like (cartilage).

Working Principle

Epithelium is classified by cell shape and layering: squamous (flat, for diffusion), cuboidal (for secretion and absorption), columnar (for absorption in the intestine), ciliated columnar (moves mucus in the respiratory tract) and glandular (secretes).

Applications

- Areolar tissue fills the space inside organs and repairs tissues.
- Adipose tissue stores fat and insulates the body against heat loss.
- Tendons attach muscle to bone; ligaments attach bone to bone.

✓ Points to Remember (Exam Focus)

- ✓ Epithelium rests on a basement membrane and has almost no intercellular space.
- ✓ Blood is a connective tissue with a fluid matrix called plasma.
- ✓ Bone matrix is hard, made of calcium and phosphorus compounds; cartilage matrix is flexible.
- ✓ Tendon connects muscle to bone (tough, less flexible); Ligament connects bone to bone (elastic, strong).
- ✓ Adipose tissue stores fat and acts as an insulator.

⚠ Common Student Mistakes

- ⚠ Refusing to accept blood as a tissue — it is a connective tissue because its cells share a common matrix and function.
- ⚠ Swapping tendon and ligament — 'tendon to the bone from the muscle'.
- ⚠ Thinking cartilage is a type of bone; it is a separate, flexible connective tissue.

**Memory Hook**

'Epithelium = the wrapping paper; Connective = the packing material' — and 'LiGament joins Bone to Bone, TenDon Ties muscle Down to bone'.

**Advanced Insights**

Blood and bone appear utterly different but share an origin — bone marrow manufactures blood cells. Their common feature, a cell-poor and matrix-rich structure derived from embryonic mesoderm, is exactly what defines connective tissue.

II Quick Recap — Topics Covered So Far

- **Plant Tissues — Meristematic and Permanent** — Plant tissues are either meristematic, which divide and cause growth, or permanent, which have lost the power to divide and perform a fixed function.
- **Animal Tissues — Epithelial and Connective** — Epithelial tissue covers and lines body surfaces, while connective tissue binds, supports and transports — blood and bone are both connective tissues.

9.3 Muscular and Nervous Tissues

 **Estimated time:** 13 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Muscular tissue produces movement by contracting, and nervous tissue carries electrical messages rapidly around the body.

Real-Life Example

Lifting a bag uses voluntary striated muscle, digesting your lunch uses involuntary smooth muscle, and your heart uses a third kind that never tires.

Detailed Explanation

Definition

Muscular tissue consists of elongated cells called muscle fibres that contract and relax to cause movement. Nervous tissue consists of neurons specialised to receive and conduct impulses.

Key Idea

There are three kinds of muscle: striated (skeletal, voluntary, cylindrical, multinucleate), smooth (unstriated, involuntary, spindle-shaped, uninucleate) and cardiac (striated but involuntary, branched, found only in the heart).

Working Principle

A neuron has a cell body with a nucleus, short branching dendrites that receive impulses, and a long axon that carries the impulse away. The impulse travels as an electrical signal and crosses to the next neuron chemically at a synapse.

Applications

- Skeletal muscles attached to bones move the limbs at will.
- Smooth muscles in the alimentary canal move food along by peristalsis.
- Cardiac muscle contracts rhythmically throughout life without fatigue.

✓ Points to Remember (Exam Focus)

- ✓ Striated = voluntary, cylindrical, multinucleate, attached to bones.
- ✓ Smooth (unstriated) = involuntary, spindle-shaped, single nucleus, in internal organs.
- ✓ Cardiac = striated but involuntary, branched, only in the heart.
- ✓ A neuron is the longest cell in the human body.
- ✓ Nervous tissue makes up the brain, spinal cord and nerves.

⚠ Common Student Mistakes

- ⚠ Calling cardiac muscle voluntary because it is striated — it is striated but involuntary.
- ⚠ Saying smooth muscles are found in the limbs; they are found in internal organs.
- ⚠ Confusing dendrite (receives) with axon (transmits).



Memory Hook

'Striated = Steered by you; Smooth = Silently automatic; Cardiac = both Striated and Silent'.



Advanced Insights

Cardiac muscle cells are joined by intercalated discs containing gap junctions, so an electrical impulse spreads through the whole heart almost instantly. This is why the heart contracts as one unit rather than fibre by fibre.

II Quick Recap — Topics Covered So Far

- Muscular and Nervous Tissues — Muscular tissue produces movement by contracting, and nervous tissue carries electrical messages rapidly around the body.

Chapter 9 — End of Chapter Revision

All Memory Hooks

- Plant Tissues — Meristematic and Permanent: 'Xylem = eXit upward with water; Phloem = Food, both ways' — and 'Para stores, Colla bends, Sclera hardens'.
- Animal Tissues — Epithelial and Connective: 'Epithelium = the wrapping paper; Connective = the packing material' — and 'LiGament joins Bone to Bone, TenDon Ties muscle Down to bone'.
- Muscular and Nervous Tissues: 'Striated = Steered by you; Smooth = Silently automatic; Cardiac = both Striated and Silent'.

All Common Mistakes

-  Saying phloem carries water — xylem carries water; phloem carries food.
-  Calling sclerenchyma a living tissue; its cells are dead at maturity.
-  Confusing apical meristem (length) with lateral meristem (girth).
-  Refusing to accept blood as a tissue — it is a connective tissue because its cells share a common matrix and function.
-  Swapping tendon and ligament — 'tendon to the bone from the muscle'.
-  Thinking cartilage is a type of bone; it is a separate, flexible connective tissue.
-  Calling cardiac muscle voluntary because it is striated — it is striated but involuntary.
-  Saying smooth muscles are found in the limbs; they are found in internal organs.
-  Confusing dendrite (receives) with axon (transmits).

Quick Revision Section

- Plant Tissues — Meristematic and Permanent: Plant tissues are either meristematic, which divide and cause growth, or permanent, which have lost the power to divide and perform a fixed function.
- Animal Tissues — Epithelial and Connective: Epithelial tissue covers and lines body surfaces, while connective tissue binds, supports and transports — blood and bone are both connective tissues.
- Muscular and Nervous Tissues: Muscular tissue produces movement by contracting, and nervous tissue carries electrical messages rapidly around the body.

Chapter 10: Diversity in Living Organisms

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Idea that living things differ greatly from one another.
- Basic difference between plants and animals.
- Knowledge that cells may or may not have a nucleus.
- Familiarity with common animals such as earthworm, fish and frog.

10.1 Classification and Binomial Nomenclature

 **Estimated time:** 13 mins **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

Quick Summary

Classification arranges organisms into groups based on similarities and differences, and binomial nomenclature gives every organism a unique two-word scientific name.

Real-Life Example

A shopkeeper arranges goods aisle by aisle so anything can be found quickly — biologists do exactly the same with about two million known species.

Detailed Explanation

Definition

Classification is the arrangement of organisms into groups on the basis of their similarities and differences. Binomial nomenclature is the system of naming an organism by two words — the genus name followed by the species name.

Key Idea

Common names differ from place to place and language to language, so a single unambiguous scientific name is needed. Carolus Linnaeus, the father of taxonomy, introduced binomial nomenclature.

Working Principle

The hierarchy runs Kingdom → Phylum (or Division in plants) → Class → Order → Family → Genus → Species. As you move down, the number of organisms in the group decreases while their similarity increases.

Applications

- *Homo sapiens* (human), *Mangifera indica* (mango), *Panthera tigris* (tiger).
- Whittaker's five-kingdom system: Monera, Protista, Fungi, Plantae, Animalia.
- The species is the basic unit of classification.

✓ Points to Remember (Exam Focus)

- ✓ Hierarchy: Kingdom, Phylum, Class, Order, Family, Genus, Species.
- ✓ Genus name begins with a capital letter, species name with a small letter; both are italicised (or underlined separately when handwritten).
- ✓ Carolus Linnaeus is the father of taxonomy and gave binomial nomenclature.
- ✓ Whittaker proposed the five-kingdom classification in 1969.
- ✓ Basis of classification: cell type (pro/eukaryotic), cell number, mode of nutrition, body organisation.

⚠ Common Student Mistakes

- ⚠ Writing the species name with a capital letter — only the genus name is capitalised.
- ⚠ Underlining the two words of a scientific name together; they must be underlined separately.
- ⚠ Reversing the hierarchy — Kingdom is the largest group, Species the smallest.

**Memory Hook**

'King Philip Came Over For Good Soup' — Kingdom, Phylum, Class, Order, Family, Genus, Species.

**Advanced Insights**

Modern classification is increasingly based on DNA sequence rather than appearance. This has revealed that some organisms placed together on the basis of looks are only distantly related — an example of convergent evolution producing similar body forms independently.

10.2 The Five Kingdoms

🕒 **Estimated time:** 15 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Whittaker's five kingdoms — Monera, Protista, Fungi, Plantae and Animalia — separate organisms by cell type, cell number and mode of nutrition.

**Real-Life Example**

A bacterium in curd, an Amoeba in pond water, a mushroom on a log, a mango tree and a tiger are one example from each of the five kingdoms.

**Detailed Explanation****Definition**

The five-kingdom classification proposed by R. H. Whittaker in 1969 groups all organisms into Monera, Protista, Fungi, Plantae and Animalia.

Key Idea

The three criteria used are cell structure (prokaryotic or eukaryotic), body organisation (unicellular or multicellular) and mode of nutrition (autotrophic or heterotrophic).

Working Principle

Monera are prokaryotic and unicellular with no true nucleus. Protista are eukaryotic and unicellular. Fungi are eukaryotic, mostly multicellular, and saprophytic with chitin cell walls. Plantae are multicellular autotrophs with cellulose cell walls. Animalia are multicellular heterotrophs without cell walls.

Applications

- Monera: bacteria, blue-green algae, mycoplasma.
- Protista: Amoeba, Paramecium, Euglena, diatoms.
- Fungi: yeast, mushroom, Penicillium, Rhizopus.

✓ Points to Remember (Exam Focus)

- ✓ Monera is the only prokaryotic kingdom — no true nucleus and no membrane-bound organelles.
- ✓ Protista are all unicellular eukaryotes and may be autotrophic or heterotrophic.
- ✓ Fungal cell walls are made of chitin, plant cell walls of cellulose.
- ✓ Plantae are autotrophic; Animalia are heterotrophic and have no cell wall.
- ✓ Euglena is a bridge organism — it photosynthesises like a plant but moves like an animal.

⚠ Common Student Mistakes

- ⚠ Placing fungi in the plant kingdom — fungi have no chlorophyll and are heterotrophic.
- ⚠ Calling all unicellular organisms Monera; unicellular eukaryotes belong to Protista.
- ⚠ Saying blue-green algae are plants; they are prokaryotes and belong to Monera.



Memory Hook

'MoProFunPlanAni' — Monera, Protista, Fungi, Plantae, Animalia, in order of increasing complexity.



Advanced Insights

The five-kingdom system has since been superseded by the three-domain system (Bacteria, Archaea, Eukarya), because Archaea — organisms of hot springs and salt lakes — turned out to be as different from bacteria as bacteria are from us.

II Quick Recap — Topics Covered So Far

- Classification and Binomial Nomenclature — Classification arranges organisms into groups based on similarities and differences, and binomial nomenclature gives every organism a unique two-word scientific name.
- The Five Kingdoms — Whittaker's five kingdoms — Monera, Protista, Fungi, Plantae and Animalia — separate organisms by cell type, cell number and mode of nutrition.

10.3 Plant and Animal Groups

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**    

Quick Summary

Plants are divided into cryptogams, which have hidden reproductive organs, and phanerogams, which bear seeds; animals are grouped by increasing body complexity from Porifera to Chordata.

Real-Life Example

Moss on a damp wall, a fern in a shaded pot, a pine cone and a mango are the four great plant groups lined up in a single garden.

Detailed Explanation

Definition

Cryptogams are seedless plants with concealed reproductive organs — Thallophyta, Bryophyta and Pteridophyta. Phanerogams bear seeds and are divided into Gymnosperms (naked seeds) and Angiosperms (seeds enclosed in fruit).

Key Idea

Plant classification moves from no body differentiation (algae) through no vascular tissue (mosses) to vascular tissue without seeds (ferns), to naked seeds and finally to flowers and fruits.

Working Principle

Animal classification tracks body organisation: cellular level (Porifera), tissue level with radial symmetry (Coelenterata), organ level with bilateral symmetry (Platyhelminthes onward), and finally a notochord and backbone (Chordata, of which Vertebrata is the major subphylum).

Applications

- Bryophyta are called the 'amphibians of the plant kingdom' as they need water for reproduction.
- Angiosperms are further divided into monocots (one cotyledon) and dicots (two cotyledons).
- Vertebrate classes: Pisces, Amphibia, Reptilia, Aves, Mammalia.

Points to Remember (Exam Focus)

- ✓ Thallophyta = algae; Bryophyta = mosses; Pteridophyta = ferns (first with vascular tissue).
- ✓ Gymnosperms bear naked seeds (pine, cycas); Angiosperms bear seeds inside fruits.

- ✓ Porifera (sponges) have pores and are the simplest multicellular animals.
- ✓ Coelenterata show radial symmetry; from Platyhelminthes onward animals are bilaterally symmetrical.
- ✓ Mammals have mammary glands, hair, and are mostly viviparous — the platypus is the egg-laying exception.

⚠ Common Student Mistakes

- ⚠ Calling algae 'plants with roots and leaves' — Thallophyta have an undifferentiated thallus.
- ⚠ Thinking gymnosperms have flowers and fruits; they have naked seeds in cones.
- ⚠ Saying all mammals give birth to live young — the platypus and echidna lay eggs.



Memory Hook

'CRYPTic = hidden reproduction; PHANEROGams = plain seeds on show' — and for plants, 'Algae, Moss, Fern, Cone, Flower' in order of advancement.



Advanced Insights

Angiosperms dominate the modern world because the flower and fruit solve two problems at once: the flower recruits animals for pollination, and the fruit recruits animals for seed dispersal — outsourcing both stages of reproduction to mobile partners.

II Quick Recap — Topics Covered So Far

- Plant and Animal Groups — Plants are divided into cryptogams, which have hidden reproductive organs, and phanerogams, which bear seeds; animals are grouped by increasing body complexity from Porifera to Chordata.

Chapter 10 — End of Chapter Revision

All Memory Hooks

- Classification and Binomial Nomenclature: 'King Philip Came Over For Good Soup' — Kingdom, Phylum, Class, Order, Family, Genus, Species.
- The Five Kingdoms: 'MoProFunPlanAni' — Monera, Protista, Fungi, Plantae, Animalia, in order of increasing complexity.
- Plant and Animal Groups: 'CRYPTic = hidden reproduction; PHANEROGams = plain seeds on show' — and for plants, 'Algae, Moss, Fern, Cone, Flower' in order of advancement.

All Common Mistakes

-  Writing the species name with a capital letter — only the genus name is capitalised.
-  Underlining the two words of a scientific name together; they must be underlined separately.
-  Reversing the hierarchy — Kingdom is the largest group, Species the smallest.
-  Placing fungi in the plant kingdom — fungi have no chlorophyll and are heterotrophic.
-  Calling all unicellular organisms Monera; unicellular eukaryotes belong to Protista.
-  Saying blue-green algae are plants; they are prokaryotes and belong to Monera.
-  Calling algae 'plants with roots and leaves' — Thallophyta have an undifferentiated thallus.
-  Thinking gymnosperms have flowers and fruits; they have naked seeds in cones.
-  Saying all mammals give birth to live young — the platypus and echidna lay eggs.

Quick Revision Section

- Classification and Binomial Nomenclature: Classification arranges organisms into groups based on similarities and differences, and binomial nomenclature gives every organism a unique two-word scientific name.
- The Five Kingdoms: Whittaker's five kingdoms — Monera, Protista, Fungi, Plantae and Animalia — separate organisms by cell type, cell number and mode of nutrition.
- Plant and Animal Groups: Plants are divided into cryptogams, which have hidden reproductive organs, and phanerogams, which bear seeds; animals are grouped by increasing body complexity from Porifera to Chordata.

Chapter 11: Genetics

Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge that the nucleus controls the cell and contains chromosomes.
- Idea that characters pass from parents to children.
- Basic understanding of sexual reproduction and gametes.
- Familiarity with the term DNA.

11.1 Chromosomes, DNA and Genes

 **Estimated time:** 14 mins **Difficulty:**  Medium **★★☆** **Exam Importance:**     

Quick Summary

Chromosomes inside the nucleus are made of DNA, and a gene is the segment of DNA that carries the instruction for one character.

Real-Life Example

The reason a child has her mother's eyes and her father's hairline is that each parent contributed one chromosome of every pair.

Detailed Explanation

Definition

A chromosome is a thread-like structure in the nucleus made of DNA and protein; DNA (deoxyribonucleic acid) is the molecule that stores hereditary information; a gene is a segment of DNA that codes for a particular character.

Key Idea

DNA is a double helix — two strands twisted around each other, discovered by Watson and Crick in 1953. The strands are held together by pairs of nitrogenous bases: adenine always pairs with thymine, and guanine always pairs with cytosine.

Working Principle

Humans have 46 chromosomes in 23 pairs. Each pair has one chromosome from the mother and one from the father, so every character is represented twice. During gamete formation the pairs separate, so each gamete carries 23 chromosomes.

Applications

- DNA fingerprinting is used in crime investigation and paternity testing.
- The Human Genome Project mapped all the genes in human DNA.
- Genetic counselling uses karyotypes to detect chromosomal disorders.

✓ Points to Remember (Exam Focus)

- ✓ Humans have 46 chromosomes = 23 pairs (22 pairs autosomes + 1 pair sex chromosomes).
- ✓ DNA is a double helix; base pairing is A–T and G–C.
- ✓ A gene is a segment of DNA that determines one character.
- ✓ Watson and Crick proposed the double-helix model of DNA in 1953.
- ✓ Gregor Mendel, the father of genetics, worked on the garden pea.

⚠ Common Student Mistakes

- ⚠ Saying humans have 46 pairs of chromosomes — they have 46 chromosomes, i.e. 23 pairs.
- ⚠ Confusing gene (a segment of DNA) with chromosome (the whole DNA-protein thread).
- ⚠ Pairing A with G — adenine always pairs with thymine, guanine with cytosine.

**Memory Hook**

'Chromosome is the book, gene is the sentence, DNA is the alphabet' — and pair 'A with T, G with C' (Apple–Tree, Green–Cabbage).

**Advanced Insights**

Only about 2% of human DNA actually codes for proteins. The rest, once dismissed as 'junk DNA', is now known to contain switches that decide when and where each gene is turned on — which is why humans and chimpanzees differ so much despite nearly identical genes.

11.2 Human Genetics — Blood Groups and Sex Determination

⌚ Estimated time: 12 mins **Difficulty:** ● Medium ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

Blood groups and sex are both determined by genes; the ABO blood groups depend on antigens on red cells, and the sex of a child depends on whether the fertilizing sperm carries X or Y.

**Real-Life Example**

Before any transfusion the hospital checks blood group and Rh factor, because giving the wrong group causes the donated cells to clump and can be fatal.

**Detailed Explanation****Definition**

Blood group is determined by the type of antigen present on the surface of red blood cells — A, B, both (AB) or neither (O). Sex determination is the mechanism by which the sex of an individual is fixed at fertilization.

Key Idea

Group O is the universal donor because its red cells carry no A or B antigen; group AB is the universal recipient because its plasma has no anti-A or anti-B antibodies. The Rh factor is a separate antigen that is either present (Rh⁺) or absent (Rh⁻).

Working Principle

The mother's ovum always carries an X chromosome. A sperm carries either X or Y. X + X gives a girl, X + Y gives a boy, so the sex of the child is determined entirely by the father's sperm and the chance is roughly 50:50.

Applications

- Blood group testing before transfusion and before surgery.
- Rh incompatibility between an Rh⁻ mother and an Rh⁺ foetus is prevented by anti-D injection.
- Genetic disorders: haemophilia and colour blindness are sex-linked; sickle-cell anaemia and thalassaemia are not.

✓ Points to Remember (Exam Focus)

- ✓ Blood groups: A, B, AB and O; O is the universal donor, AB the universal recipient.
- ✓ Female = XX, Male = XY; the ovum always carries X.
- ✓ The father determines the sex of the child.
- ✓ Haemophilia and colour blindness are X-linked and far more common in males.
- ✓ Down's syndrome is caused by an extra copy of chromosome 21.

⚠ Common Student Mistakes

- ⚠ Blaming the mother for the sex of the child — she can contribute only an X chromosome.
- ⚠ Calling AB the universal donor; AB is the universal recipient, O is the universal donor.
- ⚠ Thinking colour blindness affects males and females equally; males need only one affected X.

💡 Memory Hook

'O Offers to all, AB Accepts from all' — and 'Father's sperm flips the coin'.

🚀 Advanced Insights

Males are more affected by X-linked disorders because they have only one X: a single faulty copy has no healthy partner to mask it. Females carrying one faulty copy are usually unaffected carriers — the biological reason haemophilia was called 'the royal disease' passed silently through queens.

II Quick Recap — Topics Covered So Far

- Chromosomes, DNA and Genes — Chromosomes inside the nucleus are made of DNA, and a gene is the segment of DNA that carries the instruction for one character.
- Human Genetics — Blood Groups and Sex Determination — Blood groups and sex are both determined by genes; the ABO blood groups depend on antigens on red cells, and the sex of a child depends on whether the fertilizing sperm carries X or Y.

11.3 Genetic Disorders and Genetic Engineering

Estimated time: 12 mins **Difficulty:** Medium **★★☆** **Exam Importance:**

Quick Summary

Genetic disorders arise from faulty genes or abnormal chromosome numbers, and genetic engineering lets scientists deliberately alter genes to produce medicines and improved crops.

Real-Life Example

The insulin injected by a person with diabetes today is human insulin made by bacteria that were given the human insulin gene.

Detailed Explanation

Definition

A genetic disorder is a disease caused by an abnormality in an individual's genes or chromosomes. Genetic engineering is the deliberate modification of an organism's genetic material by inserting, deleting or altering genes.

Key Idea

Disorders may be gene-based (sickle-cell anaemia, thalassaemia, haemophilia, colour blindness) or chromosome-based (Down's syndrome from an extra chromosome 21, Turner's and Klinefelter's syndromes from abnormal sex chromosomes).

Working Principle

In recombinant DNA technology the desired gene is cut out with restriction enzymes, joined into a plasmid vector with ligase, and inserted into a bacterium. The bacterium then multiplies and manufactures the required protein.

Applications

- Human insulin, growth hormone and hepatitis-B vaccine are produced by genetically engineered bacteria.
- Bt cotton carries a bacterial gene that makes the plant resistant to bollworm.
- Gene therapy aims to replace a faulty gene with a healthy one.

Points to Remember (Exam Focus)

- ✓ Sickle-cell anaemia and thalassaemia are inherited blood disorders.

- ✓ Down's syndrome results from an extra chromosome 21 (47 chromosomes in total).
- ✓ Restriction enzymes are the 'molecular scissors' that cut DNA at specific sites.
- ✓ Genetically engineered bacteria are used to make human insulin.
- ✓ Genetically modified crops can be pest-resistant or nutritionally enriched, e.g. Golden Rice.

⚠ Common Student Mistakes

- ⚠ Calling every inherited disease a chromosomal disorder — most are single-gene disorders.
- ⚠ Saying Down's syndrome is caused by a missing chromosome; it is caused by an extra one.
- ⚠ Assuming genetic engineering and selective breeding are the same; breeding works with existing genes only.



Memory Hook

'Cut, Paste, Copy' — restriction enzyme cuts, ligase pastes, bacterium copies.



Advanced Insights

CRISPR-Cas9, adapted from a bacterial defence system, now allows a single letter of DNA to be edited precisely. It has already been used to treat sickle-cell anaemia, turning what was an incurable inherited disorder into a treatable one.

II Quick Recap — Topics Covered So Far

- Genetic Disorders and Genetic Engineering — Genetic disorders arise from faulty genes or abnormal chromosome numbers, and genetic engineering lets scientists deliberately alter genes to produce medicines and improved crops.

Chapter 11 — End of Chapter Revision

All Memory Hooks

- Chromosomes, DNA and Genes: 'Chromosome is the book, gene is the sentence, DNA is the alphabet' — and pair 'A with T, G with C' (Apple–Tree, Green–Cabbage).
- Human Genetics — Blood Groups and Sex Determination: 'O Offers to all, AB Accepts from all' — and 'Father's sperm flips the coin'.
- Genetic Disorders and Genetic Engineering: 'Cut, Paste, Copy' — restriction enzyme cuts, ligase pastes, bacterium copies.

All Common Mistakes

-  Saying humans have 46 pairs of chromosomes — they have 46 chromosomes, i.e. 23 pairs.
-  Confusing gene (a segment of DNA) with chromosome (the whole DNA-protein thread).
-  Pairing A with G — adenine always pairs with thymine, guanine with cytosine.
-  Blaming the mother for the sex of the child — she can contribute only an X chromosome.
-  Calling AB the universal donor; AB is the universal recipient, O is the universal donor.
-  Thinking colour blindness affects males and females equally; males need only one affected X.
-  Calling every inherited disease a chromosomal disorder — most are single-gene disorders.
-  Saying Down's syndrome is caused by a missing chromosome; it is caused by an extra one.
-  Assuming genetic engineering and selective breeding are the same; breeding works with existing genes only.

Quick Revision Section

- Chromosomes, DNA and Genes: Chromosomes inside the nucleus are made of DNA, and a gene is the segment of DNA that carries the instruction for one character.
- Human Genetics — Blood Groups and Sex Determination: Blood groups and sex are both determined by genes; the ABO blood groups depend on antigens on red cells, and the sex of a child depends on whether the fertilizing sperm carries X or Y.
- Genetic Disorders and Genetic Engineering: Genetic disorders arise from faulty genes or abnormal chromosome numbers, and genetic engineering lets scientists deliberately alter genes to produce medicines and improved crops.